

Financial time series

- A analysis of financial data from Planet Spiff, far far away from Tellus



Group 1

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Introduction

In the project, a dataframe *Spiff_data.csv* is provided, which consists of around 5000 daily data points for 7 time series. Each timeseries is unique and contains information described as prices from important financial markets on the planet Spiff, in a galaxy far, far away.

Each time series is presented in Figure 1, with time along the x-axis and price on the y-axis. Each timeseries spans a total time window of just below 21 years (in Earth years and thus 252 trading days), with isolated periods of missing data in shorter intervals. In Figure 1, the extreme values (values around 1000) have been removed to visualize the normal price movement of the series. These extreme values are sharp one-day spikes and will be handled later to ensure they are not forgotten.

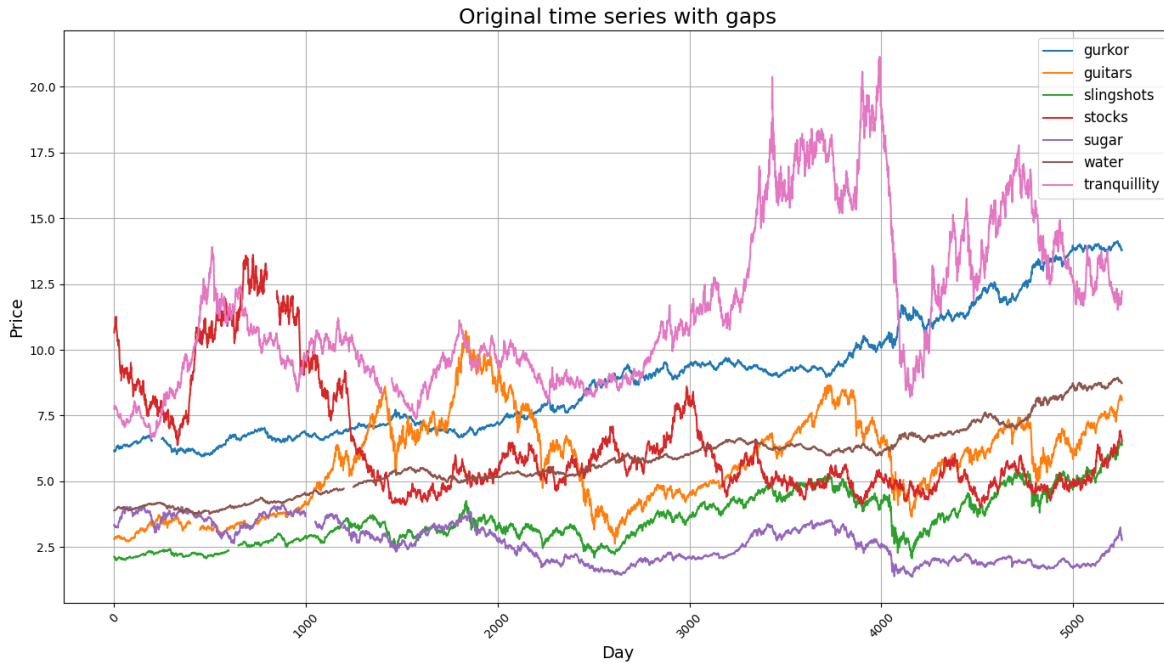


Figure 1: Original timeseries with gaps of data points.

The project is divided into four subtasks. The first part is to perform a data analysis of the different series to extract as much information as possible for each series. Information will include metrics such as seasonality, ACF, and correlation between the different series. Using this information, the next part, task 2, will identify the gaps in each series and then interpolate the gaps. This interpolation will also analyze the aspect of uncertainty.

The third part of the project was extrapolating each series by using different methods, which was intended to improve the benchmark, which is a random walk with no drift, to predict 200 additional data points.

The last part was to formulate different investment strategy and test them on the different times series. The investment strategies were evaluated through a comparison of different metrics including total return, win rate and total amount of trades.

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1 Data analysis

Firstly, when analyzing the data, one important aspect is to look at the data distribution. Here we inspect the distributions via box plots for all 7 time series in Figure 2. Visually, we can conclude that each dataset has distinct outliers at price 1000. Possibly, these points might be placeholder values or errors. To address this, we replaced each occurrence of 1000 with the average of its neighboring values, assuming this would better reflect the local trend in the data. More interestingly, though, for the box-plots of the time series Stocks, Water, and Tranquility, they also display some points as outliers close/above the 75th percentile line. Although these may be considered traditionally as outliers since they are outside the inter-quartile range [1], we can still decide to keep these values. This is because all the time series have a high kurtosis, as we see in Figure 3. In Figure 3, the kurtosis is calculated by looking at the log returns of the time series, i.e., $\log(\frac{P_t}{P_{t+1}})$ where P_t is the price of a time series at time t . According to [2], a kurtosis above 3 indicates that the distributions at hand, in our case our time series, have a heavy tail, i.e., they experience a lot of extreme events. This coincides well with the observations of Figure 2. Furthermore, we can confirm that the distributions have big tails by looking at the skewness, which is the direction where a small number of data points stretch far from the bulk of the distribution. Here we similarly see that, e.g., *stocks* have an extreme behavior, in absolute terms, the highest skewness of all the time series.

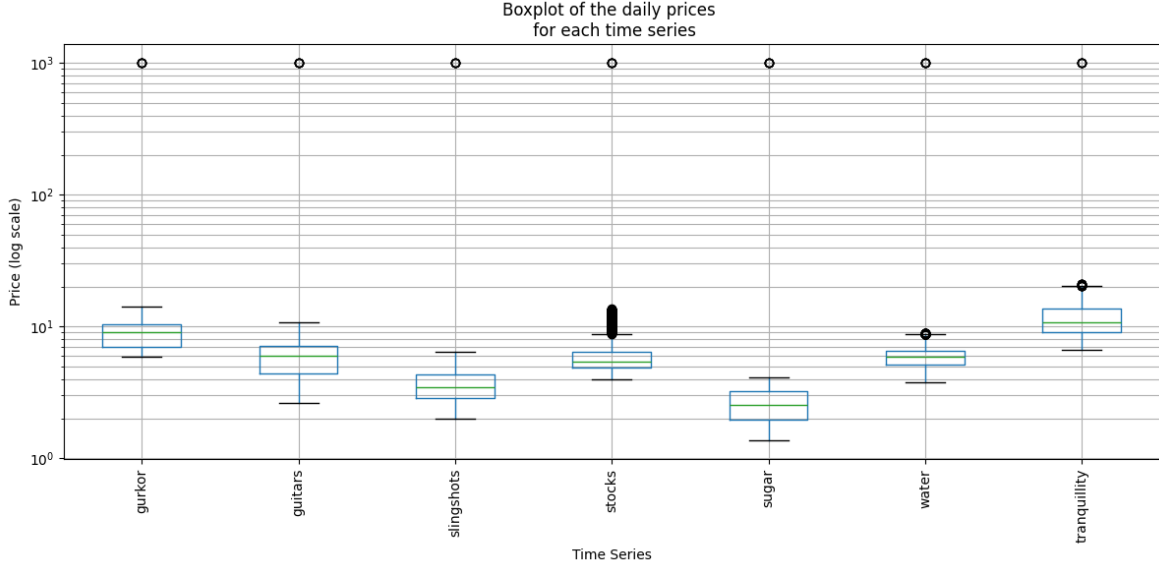


Figure 2: Boxplot of the daily prices for each timeseries on a logarithmic scale.

Furthermore, if we want to group/find relations between the time series apart from looking at skewness and kurtosis, we can also look at the correlations between the time series. In Figure 5, we can see that *slingshots* and *guitars* together show "strong" pairwise correlation, as well as *water* and *Gurkor*, which also show strong pairwise correlation. At the same time, Figure 5 also displays that the time series *Stocks* shows little to no correlation to any of the other time series. Visually, this makes sense when compared to Figure 1 that displays how the absolute price of each time series changes over time. E.g., *Slingshots* and *Guitars* seem to be somewhat correlated in Figure 1, which is in accordance with

the Pearson Correlation in Figure 5.

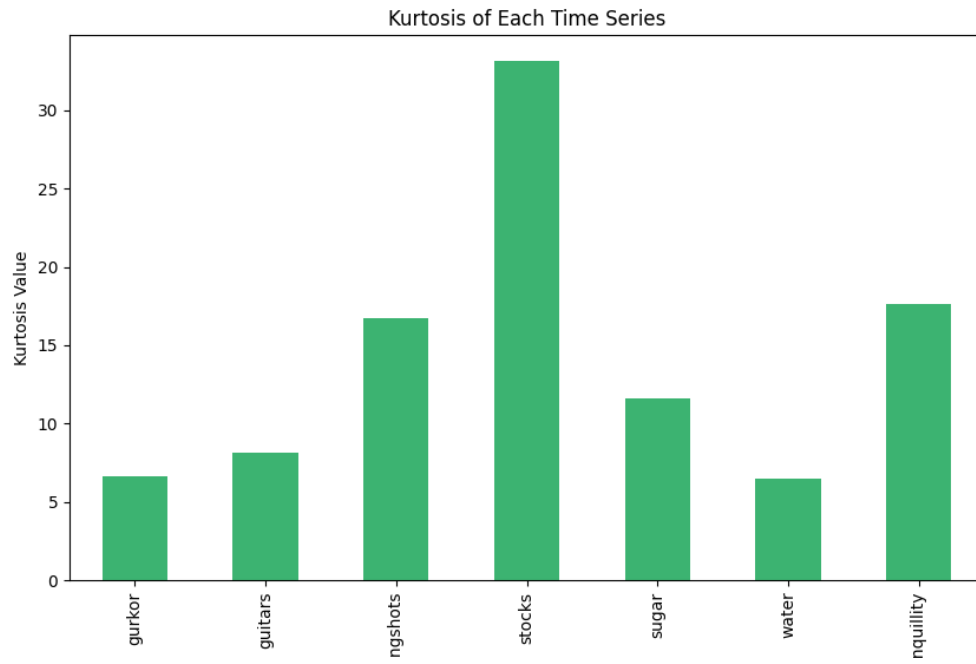


Figure 3: The Kurtosis on the log return for each timeseries.

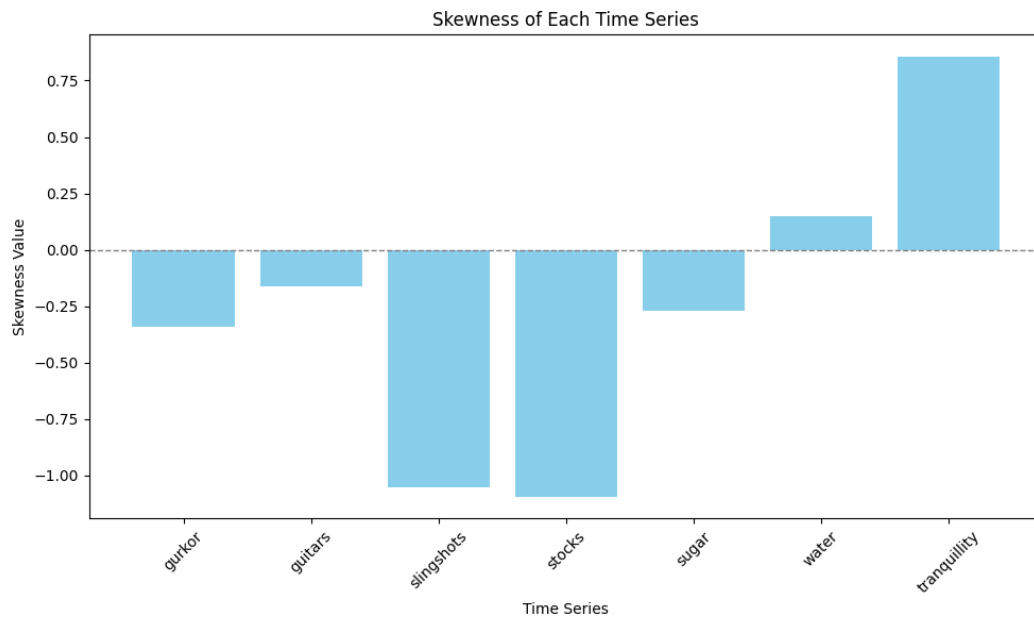


Figure 4: The skewness on the log return for each timeseries.

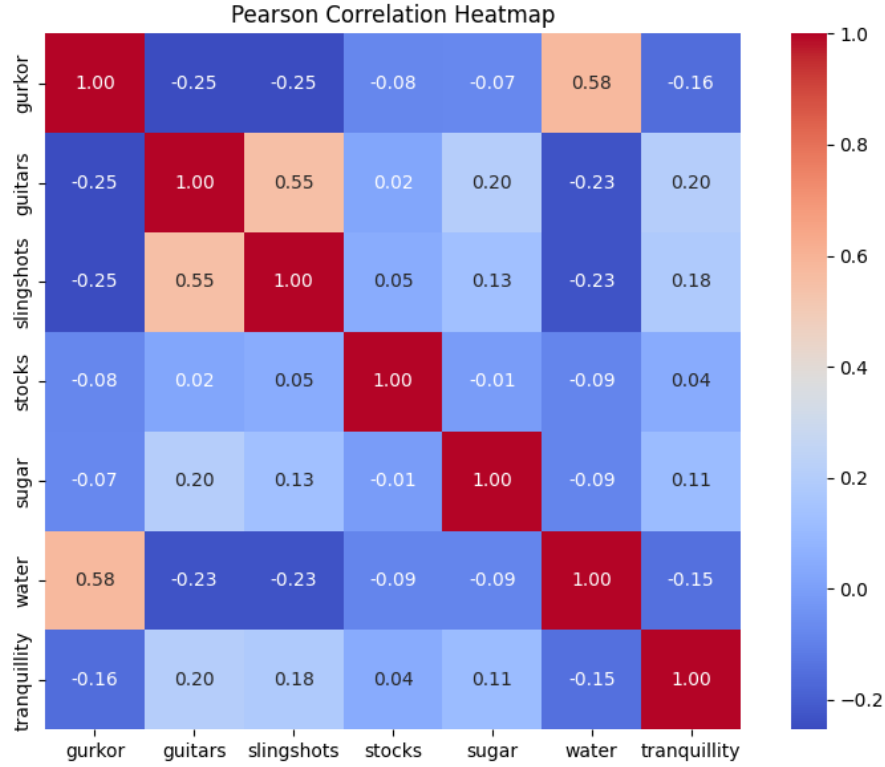


Figure 5: The Correlation between the log returns of the timeseries.

The purpose of the data analysis below is to examine the statistical properties of the time series to determine whether it is suitable for modeling and forecasting. We aim to assess the condition of stationarity for all seven series, which is a key assumption in many time series models such as ARIMA. To do this, we visually inspect both the original and transformed series alongside their autocorrelation functions (ACF:s).

In Figure 6 , the provided time series data for *stocks* is analyzed using four different panels. The price development and its corresponding ACF, the first-order log-differenced series (the log return), and its corresponding ACF. All time series have been analyzed in the same way, and the complete set of plots is provided in Appendix B

In the top-left plot, in Figure 6, we see the original price time series for *Stocks*. The series clearly shows non-stationary behavior, but without a clear seasonal pattern or overall trend. There are periods of increases and decreases in value, indicating the presence of a non-constant mean and variance. This violates the assumption of stationarity, which requires characteristics such as constant variance over time.

The top-right plot shows the ACF of the original series. There is a strong autocorrelation at low lags that slowly decays, which is typical for a non-stationary time series. This further confirms that the series is not suitable for direct use in models that assume stationarity. To address this, we applied a first-order log difference transformation to the series, displayed in the bottom-left plot. This transformation is a popular method to address skewed data and stabilize the variance. The resulting series

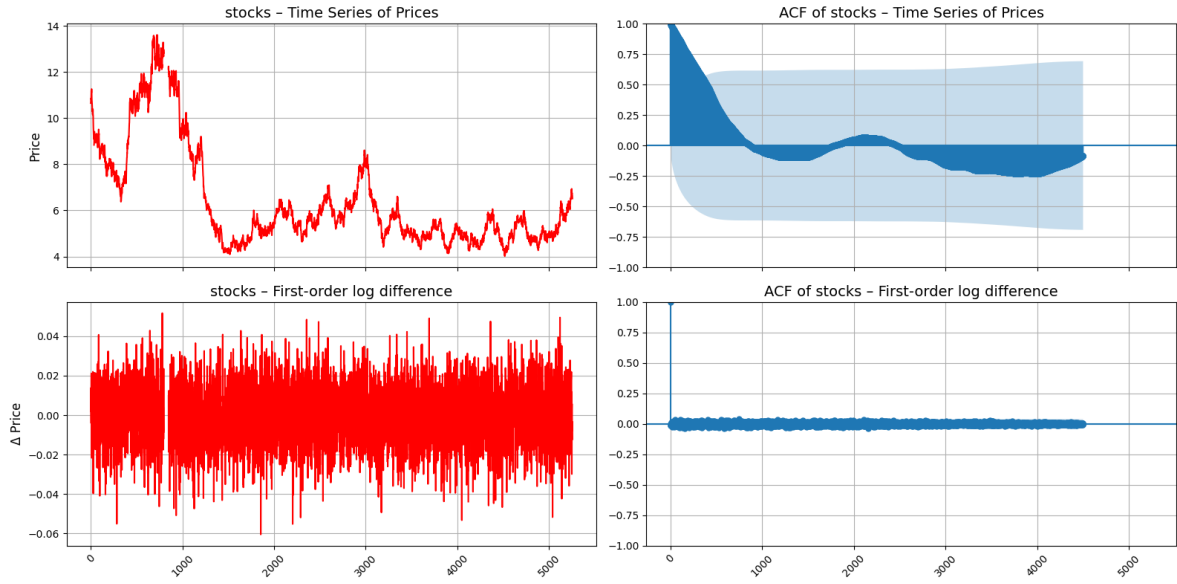


Figure 6: Time series of *stocks* with corresponding ACFs and first-order log differences.

fluctuates around zero with relatively constant variance, suggesting that it is stationary. Moreover, using the Augmented Dickey-Fuller test further validates the notion of stationarity [3].

The bottom-right plot shows the ACF of the transformed data. Here, the autocorrelation drops off quickly and remains within the confidence interval. The series now behaves more like white noise, which indicates stationarity.

This procedure is used for all the time series. All data were found to be non-stationary using a combination of visual inspection and statistical tests. In addition, no clear trend or seasonal component were discerned, which makes sense considering we are working with financial data. The choice was then done to transform the logarithm of the data by applying the differencing operator ∇ . Further evaluation of the data was then done in chapter 3, where a suitable model was chosen to represent the transformed, now stationary, data.

2 Interpolation

A significant challenge was the presence of missing data: each timeseries has a continuous gap of 50 days with no values. To interpolate these missing segments, we first analyzed the variability of the timeseries by calculating the standard deviation of daily changes (returns). This standard deviation was then used to generate a large number of random walk simulations between the known values at the boundaries of the missing segment. Only the simulated paths that ended 0.1 % from the known right-hand value were retained, ensuring consistency with the observed data.

By aggregating these valid simulations, we were able to estimate a likely trajectory for the missing values. From this ensemble, we also derived confidence intervals (68%, 95%, 99%, and 99.9%) to quantify the uncertainty of our interpolation and provide a probabilistic understanding of the data's behavior during the missing period. We also took an average value, which behaves the same way as a linear interpolation, which will be what is used for future calculations requiring one value at each index.

As for how well our method works, upon visual inspection, it behaves as expected. The obvious tells are that the uncertainty initially grows from 0 to a maximum in the middle and then back to 0 at the known value at the other end. One problem with the method, however, could be the assumption of constant volatility; if the timeseries periodically has larger returns, one could under- or overestimate the confidence interval.

To verify the effectiveness of our interpolation method, we tested it on existing complete segments of the timeseries. This validation was performed by treating known values as if they were missing, applying our interpolation procedure, and then comparing the simulated confidence intervals against the actual observed values.

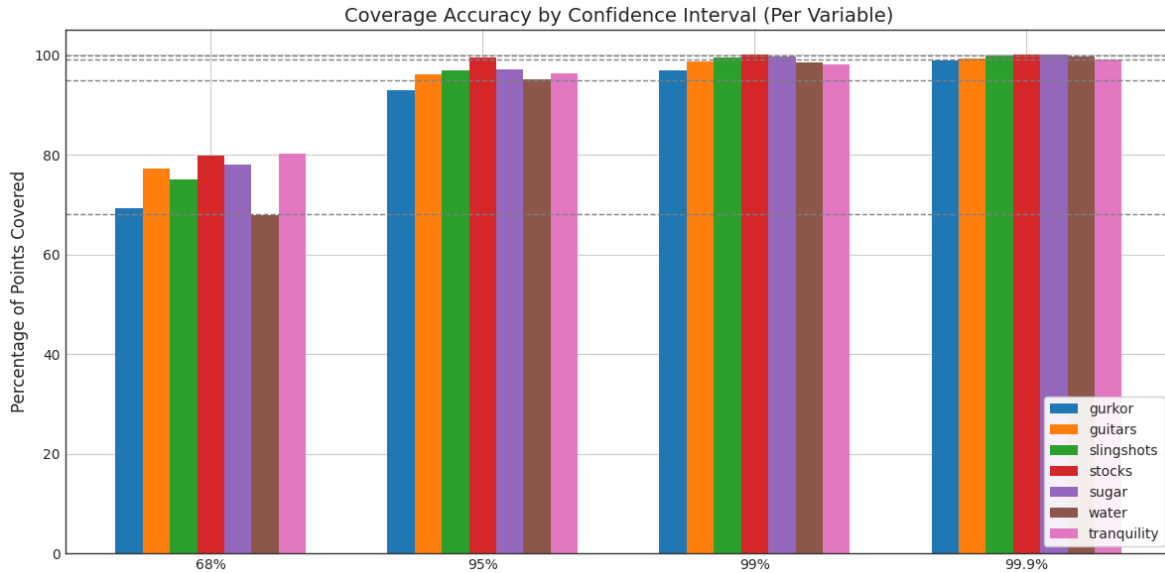


Figure 7: Validation results showing the coverage of actual data points within the predicted confidence intervals.

The results, visible in Figure 8, demonstrate that our method performs as intended in several key

aspects. The 68% confidence interval consistently contains at least 68% of the actual values, and in some timeseries captures up to 80% of points - indicating our method is slightly conservative at this confidence level. The 95% interval performs close to expectations, though with minor deviations. However, the two highest confidence intervals (99% and 99.9%) show some underestimation of the true variability. This behavior aligns with what we might expect given our assumptions: the method assumes constant volatility throughout the timeseries and normally distributed returns. In reality, if the timeseries experiences periods of higher volatility or exhibits heavier tails than a normal distribution, these highest intervals would naturally be somewhat too narrow. This could indicate that a t-distribution is worth testing.

Despite these limitations at extreme confidence levels, the method performs well for practical purposes.

Another way to improve the interpolation would be to use the timeseries correlation. As we can see from assignment 1 gurkor and water correlate, and so do guitars and slingshots. The simulation of this was not trivial as to how it was supposed to be done but we ended up with a process where it is checked how much the simulated and known timeseries correlated, if it correlated ± 0.1 from the known value it was accepted, to make the simulation faster the accepted end point was also made 0.5% instead of 0.1%. This results in graphs like the following.

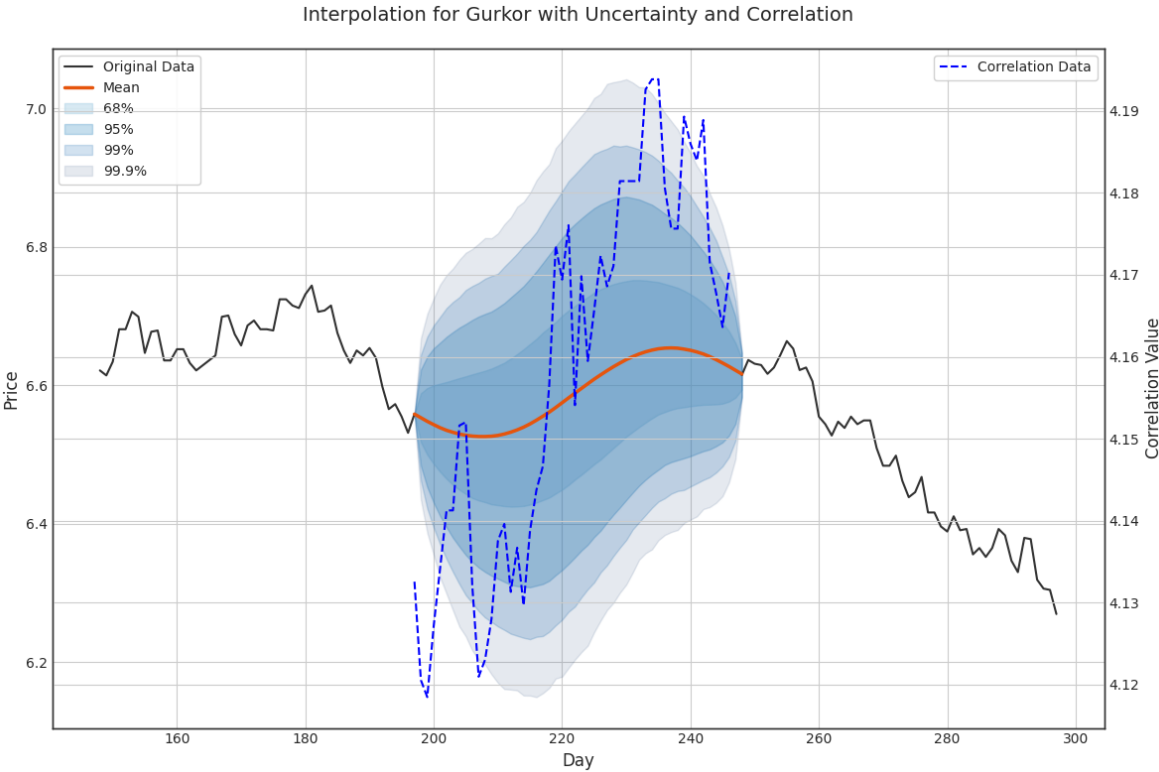


Figure 8: Validation results showing the coverage of actual data points within the predicted confidence intervals.

One problem is that we don't know that it would be correlated in this section, but this is a similar problem like before, we don't know that the volatility is the same as before, however it is a reasonable assumption.

This validation and also correlated interpolation gives us confidence that the process works well. In the rest of the report, a linear interpolation will be utilized for the non-correlated timeseries and the correlated timeseries will use the average value of the simulated values. We need to choose one value per index to simplify so we do not bring in confidence intervals into the calculations, as this would massively complicate them. This is also not a big deal as it is a small part of the data.

3 Extrapolation

In Task 3, we forecast 200 daily price points for each of the seven commodity series on planet Spiff, including uncertainty estimates. Given that Spiff’s markets are described as highly efficient, with frictionless trading and free access to information, it seems reasonable to think that predictable patterns should be rapidly traded away (eliminated) through arbitrage.

This behavior aligns closely with Earth’s ‘Efficient Market Hypothesis’ (EMH), stating that current market prices incorporate all available information [4]. Consequently, we claim that prices should, similarly to how they do on Earth, follow martingale behavior, where the best forecast of tomorrow’s price is today’s price, which can be described with the following equation:

$$\mathbb{E}[P_{t+1} \mid \mathcal{F}_t] = P_t.$$

This implies our choice of a baseline forecasting model: a random walk Without Drift, where for the whole forecast horizon, we simply predict the last known price. We define a Random walk as assuming zero expected returns and constant variance:

$$r_t = \log P_t - \log P_{t-1} = \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2).$$

Such a model would capture the nature of properly anticipated prices, due to free information and free trade, which therefore fluctuate randomly. This assumption leads to a model that serves as what we believe to be a rather solid benchmark. Indeed, previous attempts have repeatedly demonstrated the difficulty of outperforming this benchmark approach.

Finally, we note that forecasting 200 days into the future seems challenging: even under a correct model, uncertainty grows rapidly, which is inherent in a random walk process.

3.1 Methods

To see whether we can improve on that baseline, we systematically add:

- A constant drift term (eg random walk With Drift), which would imply that the expected returns are in fact not 0, but rather some non-zero value;
- ARIMA($p, 0, q$) models, with or without intercept (trend term). Due to already differencing the prices, we use 0 for the d-term;
- Volatility clustering via ARCH(1), ARCH(2), and GARCH(1,1).

Take-aways from Task 1 In Task 1, we found that first differencing each price series renders it approximately stationary (with ACF near zero), though volatility still clusters (especially for some series). Seasonality also appeared negligible, so our transformer simply takes the first differences.

Take-aways from Task 2 We use the now-interpolated data for the 7 series, from Task 2.

Rolling-origin evaluation We want every model to go through the same pipeline, to create fair estimations that can be compared against each other for evaluation. In general, to solve this task, we want to model the price forecast with a "mean model" that makes a point prediction, and model the price uncertainty through a variance model. This means that we put through the pipe fit one of each such model, which we call a pair.

All mean/variance pairs to we evaluate pass through the same pipeline:

1. At each origin t_0 (we decide to use this from 80 % of the series onward, stepping by 100 days), fit (with all available historical data)
 - the mean model on past residuals,
 - the variance model on the same residuals.
2. Forecast $h = 200$ steps ahead residuals and variances.
3. Invert the differencing to get price paths, compounded, to produce point forecasts, and variance forecasts.
4. Record metrics for point and variance forecasts to use for evaluation when comparing model pairs.

Metrics For each combo, we compute:

- *Point-forecast accuracy*: Root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), median absolute percentage error (MdAPE);
- *Variance accuracy*: $\text{MSE}_{\text{var}} = \mathbb{E}[(\hat{\sigma}^2 - (P_t - \hat{P}_t)^2)^2]$ and $\text{MAE}_{\text{var}} = \mathbb{E}\left[\left|\hat{\sigma}^2 - (P_t - \hat{P}_t)^2\right|\right]$;

Aggregation We then average each metric over all origins (to get a mean performance over the series).

3.2 Results

3.2.1 Gurkor

We begin by testing the performance difference between a random walk with or without a drift term for *gurkor*.

Table 1: Point-forecast performance

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.43630	0.30420	0.02409	0.01635
Random Walk Without Drift	0.51473	0.35258	0.02777	0.01712

We see that the model including a drift term performs better, which is expected, due to the clear upward trend of this series. We then move on to the ARIMA extension, beginning by testing all ARIMA combinations (p,0,q) of p and q, up to 3. We then also test with some higher order terms, e.g P and Q 4 and 5 (only with those combinations, in order to see what higher order terms for the

performance, and if we can clearly count these out). Below are the results of the three top-performing models:

Table 2: Top 3 performing ARIMA(p, d, q) models (constant trend, constant variance) on point-forecast metrics

Model	RMSE	MAE	MAPE	Median APE
ARIMA(3,0,3) with trend='c'	0.43513	0.30303	0.02399	0.01642
ARIMA(2,0,3) with trend='c'	0.43524	0.30226	0.02392	0.01619
ARIMA(2,0,2) with trend='c'	0.43610	0.30418	0.02410	0.01635

We see that although better than the Random Walk model With Drift, this is a very slim improvement, and we reject the idea that this model should yield a significant improvement.

We move on to the variance modeling, running our three variance models (ARCH(1), ARCH(2), and GARCH(1,1)), with our best point estimation (here, the Random Walk With Drift). The following table shows the result of such:

Table 3: Variance-forecast performance for random walk With Drift paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.17148	0.18927
ARCH(1)	0.17145	0.18922
ARCH(2)	0.17142	0.18916
GARCH(1,1)	0.17133	0.18871

We note no clear difference between the variance models. This seems reasonable, as after differencing, *Gurkor* was not one of the series with clear and different volatility clusters.

In conclusion, we decide that a Random Walk With Drift and constant variance seems to perform the best for predicting *Gurkor*. Figure 15 shows a 200-day forecast, including confidence bands:

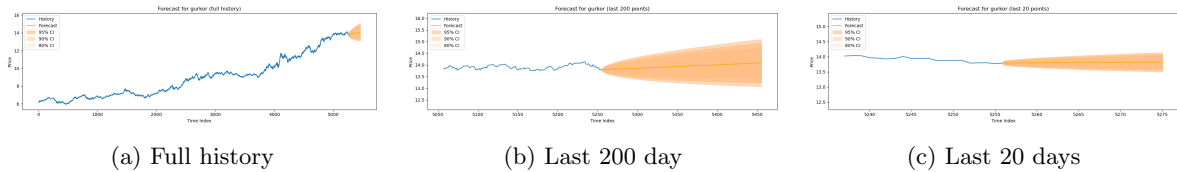


Figure 9: Gurkor price history and forecasts with different perspectives.

Below follow the remaining six commodity series, following the same structure for incremental results.

3.2.2 Guitars

Below is Random Walk Without Drift, against Random Walk With Drift.

Table 4: Point-forecast performance of the two baseline random-walk models for *Guitars*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.71969	0.55784	0.08772	0.06806
Random Walk Without Drift	0.74850	0.59762	0.09351	0.07721

Below are the top three performing ARIMA models for the series.

Table 5: Top 3 performing ARIMA(p, d, q) models on point-forecast metrics

Model	RMSE	MAE	MAPE	Median APE
ARIMA(3,0,3) trend='c'	0.71906	0.55690	0.08759	0.06761
ARIMA(2,0,2) trend='c'	0.72009	0.55906	0.08791	0.06829
ARIMA(4,0,4) trend='c'	0.72022	0.55908	0.08792	0.06813

We conclude that any ARIMA model does not yield a significant improvement for this commodity, and we move on to running a Random Walk with Drift with the variance models. Below are the variance models:

Table 6: Variance-forecast performance for RWWithDrift paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.81307	0.51166
ARCH(1)	0.81254	0.51122
ARCH(2)	0.81096	0.50995
GARCH(1,1)	0.81145	0.51052

We see that ARCH(2) gives a slightly lower MSE of the variance, but almost negligible (0.3%), and conclude that the best model for *Guitars* is Random Walk With Drift, with a constant variance. Below in Figure 10 for the 200 day forecast:

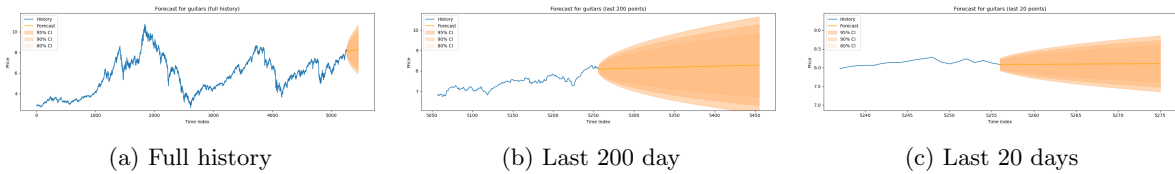


Figure 10: *Gurkor* price history and forecasts with different perspectives.

3.2.3 Slingshots

Below is Random Walk Without Drift, against Random Walk With Drift for *Slingshots*:

Table 7: Point-forecast performance of the two baseline random-walk models for *Slingshots*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.52549	0.41801	0.08991	0.07592
Random Walk Without Drift	0.55158	0.44227	0.09481	0.08338

Below are the top three performing ARIMA models for the series.

Table 8: Top 3 performing ARIMA(p, d, q) models (constant trend, constant variance) on point-forecast metrics

Model	RMSE	MAE	MAPE	Median APE
ARIMA(3,0,3) trend='c	0.52576	0.41788	0.08995	0.07559
ARIMA(1,0,3) trend='c	0.52583	0.41794	0.08997	0.07560
ARIMA(3,0,2) trend='c	0.52623	0.41826	0.09004	0.07580

We note that no ARIMA model even beats the baseline Random Walk, and move on to the variance models:

Table 9: Variance-forecast performance for RWWithDrift paired with various volatility models (second dataset)

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.21741	0.27392
ARCH(1)	0.21737	0.27386
ARCH(2)	0.21733	0.27379
GARCH(1,1)	0.21529	0.27124

We note that with each ARCH(1), ARCH(2) and GARCH(1,1), in that order, we achieve a slightly better MSE of the variance, with GARCH(1,1) being about 1% better than the constant variance. We conclude that the best model to forecast *Slingshots* is a Random Walk with Drift with GARCH(1,1). Below (in figure ??) is a plot of 200 day forecast with that model combination:

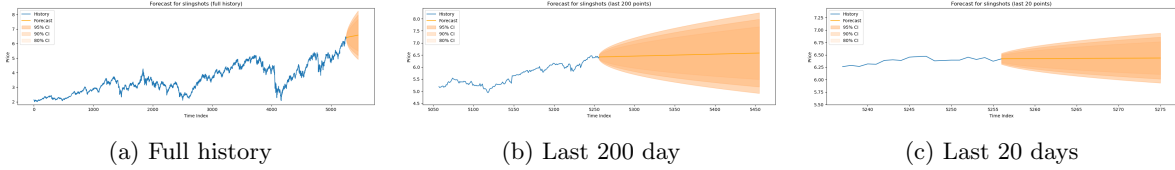


Figure 11: *Gurkor* price history and forecasts with different perspectives.

3.2.4 Stocks

Below is Random Walk Without Drift, against Random Walk With Drift for *Stocks*:

Table 10: Point-forecast performance of the two baseline random-walk models for *Stocks*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.66292	0.50402	0.09638	0.07879
Random Walk Without Drift	0.60082	0.45711	0.08849	0.06861

Below are the top three performing ARIMA models for the series.

Table 11: Top 3 performing ARIMA(p, d, q) models (constant trend, constant variance) on point-forecast metrics

Model	RMSE	MAE	MAPE	Median APE
ARIMA(1,0,1) trend='c'	0.60091	0.45716	0.08850	0.06865
ARIMA(2,0,2) trend='c'	0.60113	0.45729	0.08855	0.06847
ARIMA(3,0,3) trend='c'	0.60123	0.45746	0.08858	0.06838

We note that no ARIMA model even beats the baseline Random Walk, and move on to the variance models:

Table 12: Variance-forecast performance for RWNoDrift paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.35574	0.35368
ARCH(1)	0.35565	0.35361
ARCH(2)	0.35548	0.35347
GARCH(1,1)	0.35835	0.35588

We see almost identical behavior for the 4 variance models, and conclude that choosing anything but constant variance model likely does not yield better performance.

Below (see figure 12) is a Random Walk without Drift with Constant Variance:

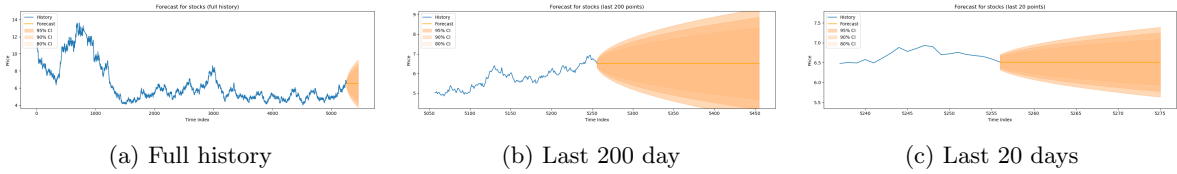


Figure 12: *Gurkor* price history and forecasts with different perspectives.

3.2.5 Sugar

Below is Random Walk Without Drift, against Random Walk With Drift for *Sugar*:

Table 13: Point-forecast performance of the two baseline random-walk models for *sugar*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.18076	0.12984	0.06480	0.05225
Random Walk Without Drift	0.16788	0.12235	0.06162	0.04975

Below are the top three performing ARIMA models for the series.

Table 14: Point-forecast performance of the top 3 ARIMA models for *sugar*

Model	RMSE	MAE	MAPE	Median APE
ARIMA(5,0,4) trend='c'	0.16780	0.12218	0.06151	0.04973
ARIMA(3,0,2) trend='c'	0.16815	0.12252	0.06169	0.04974
ARIMA(4,0,5) trend='c'	0.16823	0.12262	0.06173	0.05000

None of the models clearly beats the baseline Random Walk With Drift, and we move on to the variance models:

Table 15: Variance-forecast performance for RWNoDrift paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.00553	0.02709
ARCH(1)	0.00553	0.02709
ARCH(2)	0.00553	0.02709
GARCH(1,1)	0.00551	0.02705

We conclude that the different variance models no clear improvement over the constant variance (for the best one, GARCH(1,1) this is an improvement of 0.4%). We conclude that the best model for forecasting *Sugar* is a Random Walk with Constant Variance. The following (figure 13) is a plot for such:

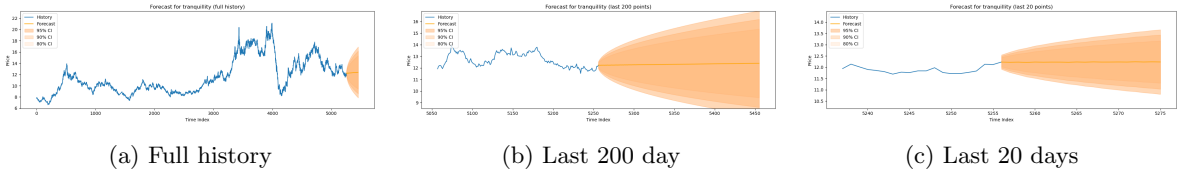


Figure 13: *Gurkor* price history and forecasts with different perspectives.

3.2.6 Tranquillity

Below is Random Walk Without Drift, against Random Walk With Drift for *Tranquillity*:

Table 16: Point-forecast performance of the two baseline random-walk models for *tranquillity*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	2.00941	1.65313	0.11863	0.10392
Random Walk Without Drift	2.00201	1.64995	0.11799	0.10466

Below are the top three performing ARIMA models for the series.

Table 17: Point-forecast performance of the top 3 ARIMA models for *tranquillity*

Model	RMSE	MAE	MAPE	Median APE
ARIMA(5,0,5) trend='c'	1.96596	1.61111	0.11573	0.09999
ARIMA(3,0,2) trend='c'	1.99870	1.64629	0.11771	0.10432
ARIMA(2,0,1) trend='c'	1.99955	1.64745	0.11779	0.10467

Here, we actually find that an ARIMA(5,0,5) with the differenced series performs what we would perform with some significance (2%).

We move on to the variances:

Table 18: Variance-forecast performance for ARIMA(5,5) with constant trend paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	36.4119	3.8417
ARCH(1)	36.3909	3.8392
ARCH(2)	36.3508	3.8347
GARCH(1,1)	36.2381	3.8221

We see an improvement using ARCH(1), ARCH(2) and GARCH(1,1) (in increasingly useful order), but the gain is still slim, with a performance increase of 0.5% for the MSE of the variance. We do not consider this significant, and conclude that the best model combination for *Tranquillity* is a Random Walk without Drift with a constant variance assumption. Below is a plot for such a 200 forecast prediction (see figure 14):

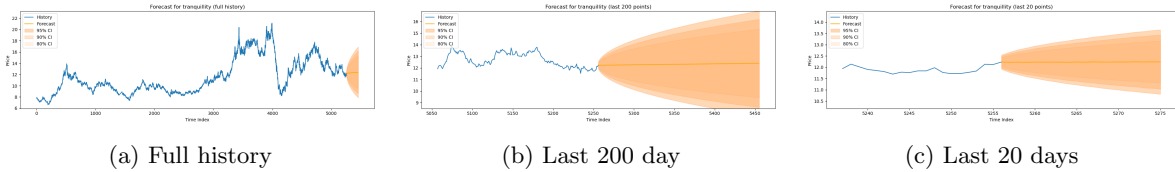


Figure 14: *Gurkor* price history and forecasts with different perspectives.

3.2.7 Water

Below is Random Walk Without Drift, against Random Walk With Drift for *Water*:

Table 19: Point-forecast performance of the two baseline random-walk models for *water*

Model	RMSE	MAE	MAPE	Median APE
Random Walk With Drift	0.33690	0.24729	0.03191	0.02462
Random Walk Without Drift	0.38576	0.28196	0.03639	0.02779

Below are the top three performing ARIMA models for the series.

Table 20: Point-forecast performance of the top 3 ARIMA models for *water*

Model	RMSE	MAE	MAPE	Median APE
ARIMA(5,0,4) trend='c'	0.33458	0.24300	0.03131	0.02319
ARIMA(5,0,5) trend='c'	0.33532	0.24496	0.03161	0.02402
ARIMA(4,0,4) trend='c'	0.33573	0.24532	0.03165	0.02422

We see an improvement of about 0.7%, but do not consider this an improvement enough to move away from using a Random Walk with Drift for *Water*.

We move on to the variance models, using with a Random Walk with Drift:

Table 21: Variance forecast performance for Random Walk paired with various volatility models

Volatility Model	MSE _{var}	MAE _{var}
Constant Variance	0.06073	0.11317
ARCH(1)	0.06073	0.11316
ARCH(2)	0.06073	0.11316
GARCH(1,1)	0.06067	0.11278

We conclude that for *Water*, the ARCH models do not seem to offer any performance increase. This seems reasonable, as *Water* is not one of the time-series that seems to exhibit high volatility clustering. Below is a day 200 forecast for *Water*:

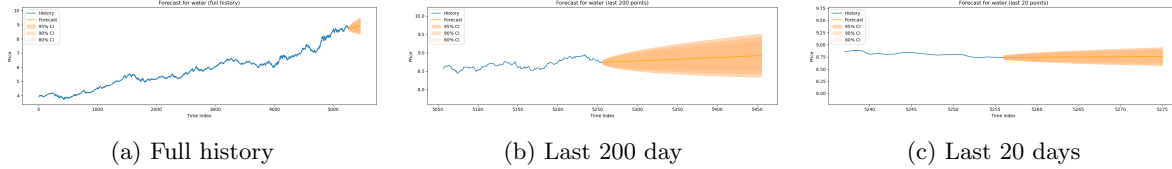


Figure 15: *Gurkor* price history and forecasts with different perspectives.

3.3 Summary of chosen models

We wrap up our findings with the following table, indicating our model pair of choice for each commodity:

Table 22: Selected forecasting models for each commodity

Commodity	Selected Model
Gurkor	Random Walk with Drift + Constant Variance
Guitars	Random Walk with Drift + Constant Variance
Slingshots	Random Walk with Drift + GARCH(1,1)
Stocks	Random Walk without Drift + Constant Variance
Sugar	Random Walk without Drift + Constant Variance
Tranquillity	ARIMA(5,0,5) with intercept + Constant Variance
Water	Random Walk with Drift + Constant Variance

3.3.1 Conclusion

Our results suggest that forecasting financial asset prices over longer horizons remains a difficult task. A natural next step is to test more sophisticated modeling techniques. For example, one might consider nonlinear approaches, machine learning methods, or regime-switching models.

4 Investment Strategies

The final task of this project was to test different investment strategies and evaluate the outcome of each strategy by comparing it with different measurements. The strategies explored and evaluated in this project are methods that are known and are commonly used in technical analysis. The strategies used are shown in table 23.

	Strategy
1	Buy and Hold
2	Moving Average Crossover (MA Cross 9 / 26)
3	Buy the Dip
4	Relative Strength Index (RSI)

Table 23: List of strategies implemented in this project.

The first and simplest strategy is "Buy and Hold." In this case, it involves purchasing the asset on day one and selling it on the final day (day 5455). The rationale behind this strategy is that, over time, markets generally trend upward despite short-term volatility and fluctuations. Since it requires no active trading or position adjustments, Buy and Hold is considered a low-effort strategy. When assessing more active or market-timing strategies, it is useful to compare their performance against this baseline approach.

The second strategy, Buying the Dips, involves identifying significant price drops—known as "dips"—that may present buying opportunities. The key challenge with this approach is accurately defining what constitutes a dip and avoiding false bottoms. In this implementation, a dip is defined as a price decline over three consecutive days. A purchase is made on the fourth day, and the position is sold on the fifth day.

The third strategy, Moving Average Crossover, involves two moving averages calculated over two different time periods, a short-term and a long-term average. A golden cross occurs when the short-term moving average crosses above the long-term moving average, signaling a potential upward trend. When the short-term calculated average crosses below the long-term, it is interpreted as a sell signal, indicating a possible downward trend.

The last method is a strategy that uses the Relative Strength Index (RSI) as an indicator to determine when the market is potentially oversold or overbought, thereby generating buy or sell signals. These signals are based on calculated RSI values, which are derived by first defining gains and losses as follows:

$$\text{Gain}_i = \begin{cases} \Delta P_i, & \Delta P_i > 0 \\ 0, & \text{else} \end{cases} \quad \text{Loss}_i = \begin{cases} |\Delta P_i|, & \Delta P_i < 0 \\ 0, & \text{else} \end{cases}$$

Then the average gain and the average loss are calculated over fourteen days through the formula:

$$\text{AvgGain} = \frac{1}{n} \sum_{i=1}^n \text{Gain}_i \quad \text{AvgLoss} = \frac{1}{n} \sum_{i=1}^n \text{Loss}_i$$

The Relative Strength (RS) is then computed, which is used to calculate the RSI:

$$RS = \frac{\text{AvgGain}}{\text{AvgLoss}} \quad RSI = 100 - \frac{100}{1+RS}$$

A buy signal is then triggered when the RSI falls below 30, indicating an oversold condition, while a sell signal occurs when the RSI exceeds 70, indicating an overbought market.

To evaluate each strategy, several performance metrics have been employed to analyze and compare their effectiveness. The investment strategies are visualized in figures over the same period to get a grasp of how they function. These metrics provide a standardized basis for assessment and facilitate meaningful comparisons between strategies. These metrics will be measured in percentage and are the following presented in table 24.

Table 24: Performance Metrics Description

Metric	Description
Total return (%)	The overall percentage gain or loss over the testing period, i.e., over the entire time series.
Annualized Volatility (%)	Presents the average volatility rate per day 252.
Sharpe Ratio	A measure of risk-adjusted return showing how much excess return is earned per risk unit.
Total amount of Trades	Includes both buying and selling trades accumulating to the total amount of trades.
Best trade (%)	The trade that amounted to the highest return.
Worst trade (%)	The trade that amounted to the greatest loss.
Average holding period	Displays the average amount of days that the method holds a position.
Win rate (%)	Percentage of trades that obtain a profit, i.e., positive outcome of the trade. Indicates the consistency of the strategy.

4.1 Strategy: Buy and Hold

The Buy and Hold strategy only has two trades in each of the assets. As a reference to the signals, the price development of the asset class, *Gurkor*, is shown in Figure 16.



Figure 16: Price and buy/sell signals.

The results for all assets are presented in the table 25.

Table 25: Performance of the Buy and Hold strategy.

Strategy	Asset	Total return (%)	Volatility(%)	Sharpe ratio	Win rate (%)	Best(%)	Worst(%)	Avg. Hold	Trades
Buy & Hold	gurkor	124.09	0.00	inf	100.00	124.09	124.09	5256.00	2
Buy & Hold	guitars	189.34	0.00	inf	100.00	189.34	189.34	5256.00	2
Buy & Hold	slingshots	200.44	0.00	inf	100.00	200.44	200.44	5256.00	2
Buy & Hold	stocks	-38.87	0.00	-inf	0.00	-38.87	-38.87	5256.00	2
Buy & Hold	sugar	-16.97	0.00	-inf	0.00	-16.97	-16.97	5256.00	2
Buy & Hold	water	124.29	0.00	inf	100.00	124.29	124.29	5256.00	2
Buy & Hold	tranquillity	57.70	0.00	inf	100.00	57.70	57.70	5256.00	2
Average		91.43	0.00	—	71.43	91.43	91.43	5256.00	2.00

Considering the total return Slingshots had the best return. On the contrary *Stocks* had a striking negative return since the asset is less in the end compared to the start. But at the end, buy and hold only depends on that the asset will increase in price, this strategy is therefore more or less a method hoping that the price will increase. Since the strategy yields only a single return value, there is no variation in returns resulting in a standard deviation equal to zero. Therefore, the Sharpe ratio may be considered misleading in this case.

4.2 Strategy: Moving Average Crossover (MA Cross)

In Figure 17, the buy and sell patterns are displayed where green and red triangles indicate investment positions. The underlying moving average lines, MA9 and MA26, represent the average return over the last 9 and 26 days, respectively. The intersection between these two lines marks the trading signals used in the strategy. The logic behind this strategy is that short-term price movements crossing long-term trends can signal a shift in the market.

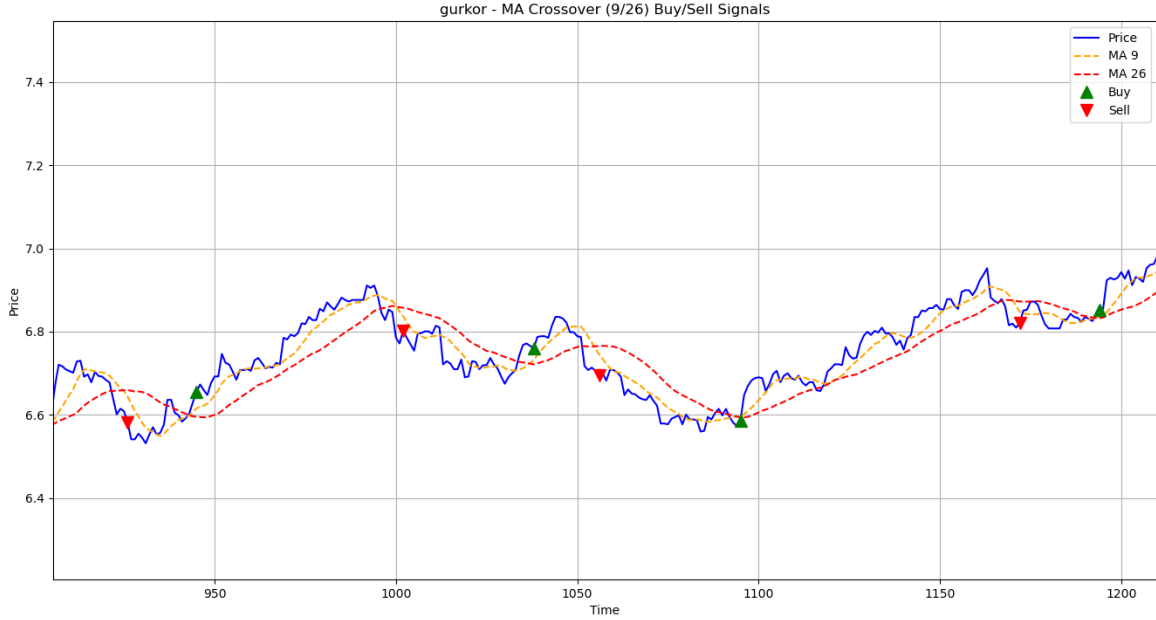


Figure 17: Price and buy/sell signals with underlying moving average (9 & 26) lines .

The result from this strategy for each asset is presented in table 26.

Table 26: Performance of the Moving Average Crossover strategy.

Strategy	Asset	Total return (%)	Volatility(%)	Sharpe ratio	Win rate (%)	Best(%)	Worst(%)	Avg. Hold	Trades
MA Cross	gurkor	87.74	4.70	0.028	57.73	0.76	-0.90	30.39	97
MA Cross	guitars	363.39	14.77	0.368	45.36	4.38	-3.84	32.02	97
MA Cross	slingshots	238.37	14.51	0.269	40.20	7.91	-8.94	30.04	102
MA Cross	stocks	-65.87	16.67	-0.406	48.39	3.42	-5.18	20.77	124
MA Cross	sugar	76.71	14.19	0.052	51.58	4.08	-4.95	25.35	95
MA Cross	water	54.66	4.12	-0.200	51.06	0.67	-0.84	31.48	94
MA Cross	tranquillity	22.79	14.12	-0.072	50.88	6.16	-3.11	21.72	114
Average		111.54	11.44	0.0056	49.89	3.63	-3.97	27.97	103.29

As seen in the table 26, the MA crossover strategy is a rather good strategy, with a total return reaching from -65.87% to 363.39 % and thus averaging a total return of 111.54% over a 20 year period. The average win rate for this strategy is 49.89% implying that almost half of each trade goes with a profit.

4.3 Strategy: Buy the Dip

In Figure 18 strategy *Buy the Dip* is shown over a short time window, showing both buy and sell signals. As seen in Figure 18 the green buy markers appear shortly after price dips.



Figure 18: Price and buy/sell signals.

The result from this strategy for each asset is presented in table 27.

Table 27: Performance of the Buy the Dip strategy.

Strategy	Asset	Total return (%)	Volatility(%)	Sharpe ratio	Win rate (%)	Best(%)	Worst(%)	Avg. Hold	Trades
Buy the Dip	gurkor	36.87	3.12	-0.463	55.79	1.20	-2.17	5.18	432
Buy the Dip	guitars	243.96	14.33	0.275	55.99	13.04	-8.89	5.36	484
Buy the Dip	slingshots	106.90	12.89	0.102	55.19	9.69	-9.13	5.16	424
Buy the Dip	stocks	-8.26	12.89	-0.200	46.48	3.96	-5.36	5.75	682
Buy the Dip	sugar	48.27	13.01	-0.021	51.47	8.12	-10.89	5.16	408
Buy the Dip	water	20.80	2.72	-0.757	52.70	1.19	-1.32	5.32	444
Buy the Dip	tranquillity	-7.50	10.08	-0.284	53.57	4.45	-5.99	5.32	504
Average		62.72	9.72	-0.193	53.60	5.95	-6.11	5.32	468.29

Buy the dip strategy is the most active strategy, conducting the most amount of trades. This strategy varied a bit when it comes to the return of the trades, reaching from -8.26% to 243.96 %. The win rate of this strategy shows that most of the trades are profitable for almost every asset.

4.4 Strategy: Relative Strength Index (RSI)

In Figure 19, the strategy *RSI* is shown over a short time window, showing both buy and sell signals. As seen in Figure 19, the green buy markers appear when the thresholds for an oversold asset are met (when the RSI is below 30), and a red sell marker appears when the RSI is above 70, indicating that the asset is overbought and thus creating a sell signal. The RSI values are shown in Figure 20, including the thresholds.

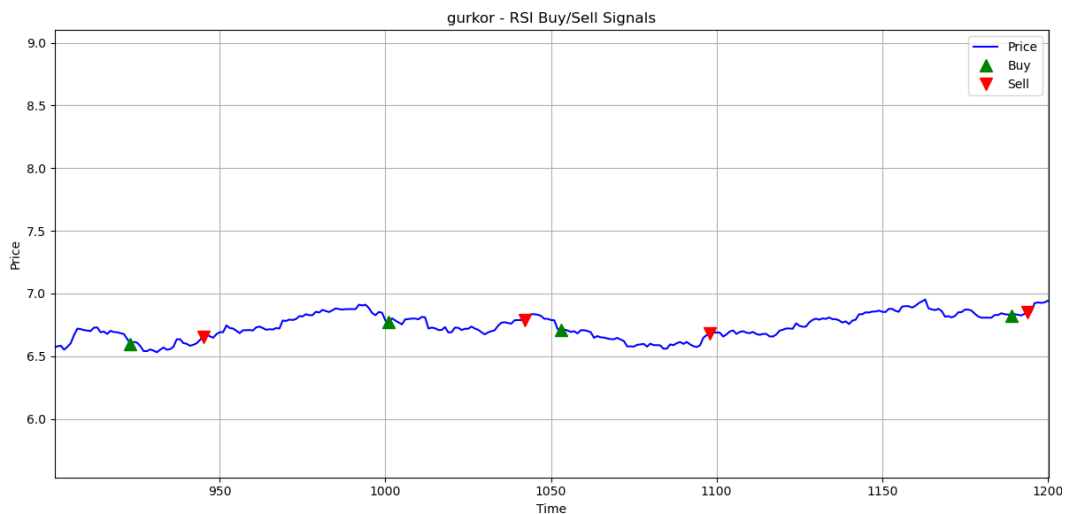


Figure 19: Price and buy/sell signals for RSI.

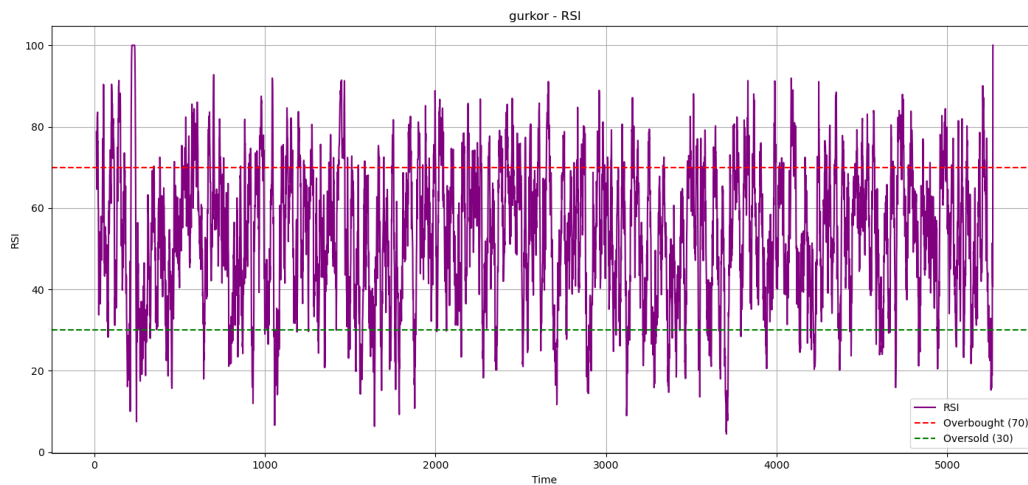


Figure 20: RSI signals, if the RSI is over the red line, the asset is overbought creating a sell signal. If the RSI is below the green line the asset is oversold indicating a buy signal.

The result from this strategy for each asset is presented in Table 28.

Table 28: Performance of the RSI strategy.

Strategy	Asset	Total return (%)	Volatility(%)	Sharpe ratio	Win rate (%)	Best(%)	Worst(%)	Avg. Hold	Trades
RSI	gurkor	23.51	3.81	-0.502	58.49	0.88	-0.83	33.77	53
RSI	guitars	-19.69	18.35	-0.129	54.00	2.67	-3.53	37.71	50
RSI	slingshots	32.46	17.90	-0.002	44.00	4.10	-9.13	37.11	50
RSI	stocks	8.46	15.95	-0.084	60.00	3.20	-3.45	43.98	55
RSI	sugar	14.97	19.53	-0.022	59.68	4.24	-8.46	40.94	62
RSI	water	29.74	3.55	-0.475	50.00	0.93	-0.68	33.20	54
RSI	tranquillity	-20.82	13.41	-0.240	57.89	4.45	-2.39	44.18	57
–	Average	9.66	13.21	-0.208	54.86	2.93	-4.64	38.27	54.43

Overall, we can observe that the RSI strategy over all assets performed worse than the risk-free investment option since the Sharpe ratio is negative. The trades over all assets are quite few but with a pretty high win-rate in general. No individual trade seemed to have lead to a big loss, therefore the RSI method doesn't indicate a good result but not a terrible either.

4.5 Summary of the performance of the strategies

To conclude the performance of all the strategies, the top five total returns obtained are displayed in table 29.

Table 29: Top 5 Best Performing Strategies (by Total Return)

Strategy	Asset	Total return (%)	Volatility (%)	Sharpe ratio	Win rate(%)	Best (%)	Worst (%)	Avg. Hold	Trades
MA Cross	guitars	363.39	14.77	0.368	45.36	4.38	-3.84	32.02	97
Buy the Dip	guitars	243.96	14.33	0.275	55.99	13.04	-8.89	5.36	484
MA Cross	slingshots	238.37	14.51	0.269	40.20	7.91	-8.94	30.04	102
Buy & Hold	slingshots	200.44	0.00	inf	100.00	200.44	200.44	5256.00	2
Buy & Hold	guitars	189.34	0.00	inf	100.00	189.34	189.34	5256.00	2

Table 30: Bottom 5 Worst Performing Strategies (by Total Return)

Strategy	Asset	Total return (%)	Volatility (%)	Sharpe ratio	Win rate (%)	Best (%)	Worst (%)	Avg. Hold	Trades
MA Cross	stocks	-65.87	16.67	-0.406	48.39	3.42	-5.18	20.77	124
Buy & Hold	stocks	-38.87	0.00	-inf	0.00	-38.87	-38.87	5256.00	2
RSI	tranquillity	-20.82	13.41	-0.240	57.89	4.45	-2.39	44.18	57
RSI	guitars	-19.69	18.35	-0.129	54.00	2.67	-3.53	37.71	50
Buy & Hold	sugar	-16.97	0.00	-inf	0.00	-16.97	-16.97	5256.00	2

It is apparent that the underlying asset is a major factor when it comes to performance. *Guitars* and *slingshot* both have a great return when held from day 1 until day 5255. But overall, it is apparent that the worst strategy is RSI since all assets result in a low average return over all the assets. According to table 30, which presents the five worst investments, RSI ends up on two of them, implying that RSI might not be the preferred one when constructing a portfolio. On the other hand, the best-performing strategy is the MA crossover with a short moving average indicator of nine days and a long moving average of twenty-six days, resulting in an average return of 111.54 %. Over 20 years, this implies a compounded annual growth of 3.66 %, which is barely better than the risk-free asset rate of 3% (total return of 85.21%). In two of the five cases, MA shows the best total return. We can thus conclude that MA crossover has the best average total return and thus beats the others. But the risk over return

relation would have to be taken into account when choosing whether to use this strategy or just invest in the risk-free asset. The potential upside of a total return of 363.39 % could be worth the risk but as always it is important to diversify and hence this project concludes that the best investment strategy on planet spiff would be to combine both a MA crossover and a Buy and Hold strategy together with risk free asset to create a nice portfolio.

References

- [1] D. S. Moore, G. P. McCabe, and B. A. Craig, *Introduction to the Practice of Statistics*, 9th ed. New York: W. H. Freeman and Company, 2017.
- [2] G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed. Pacific Grove, CA: Duxbury, 2002.
- [3] A. Lang and A. Petersson, “Financial time series,” 2022, lecture notes, version August 30, 2022, Chalmers University of Technology and University of Gothenburg.
- [4] E. F. Fama, “Efficient capital markets,” *Journal of finance*, vol. 25, no. 2, pp. 383–417, 1970.

A Appendix

B Appendix - Data analysis

This appendix presents plots for seven time series, including the price development, autocorrelation function (ACF), first-order log-differenced series, and corresponding ACF:s. Each set of four plots provides an overview of the raw and transformed series to assess stationarity.

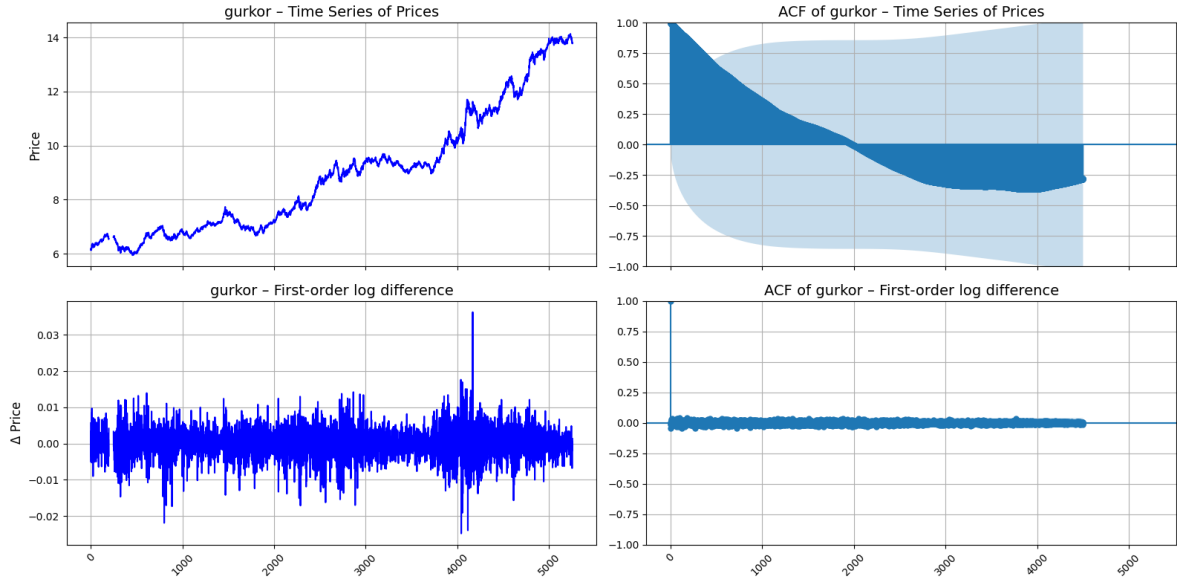


Figure 21: Time series of *gurkor* with corresponding ACFs and first-order log differences.

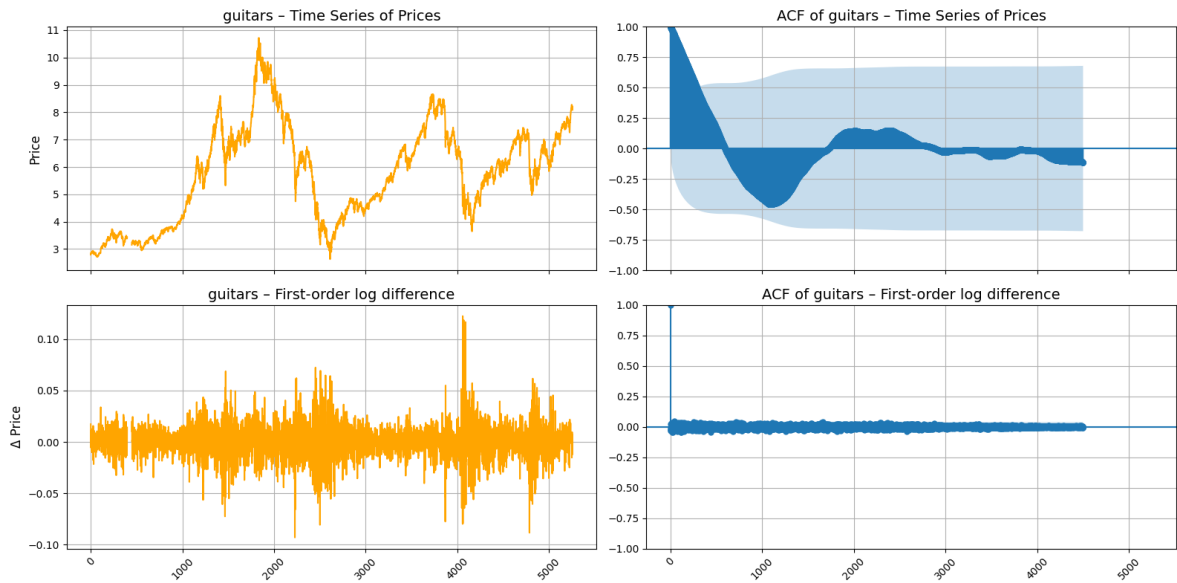


Figure 22: Time series of *guitars* with corresponding ACFs and first-order log differences.

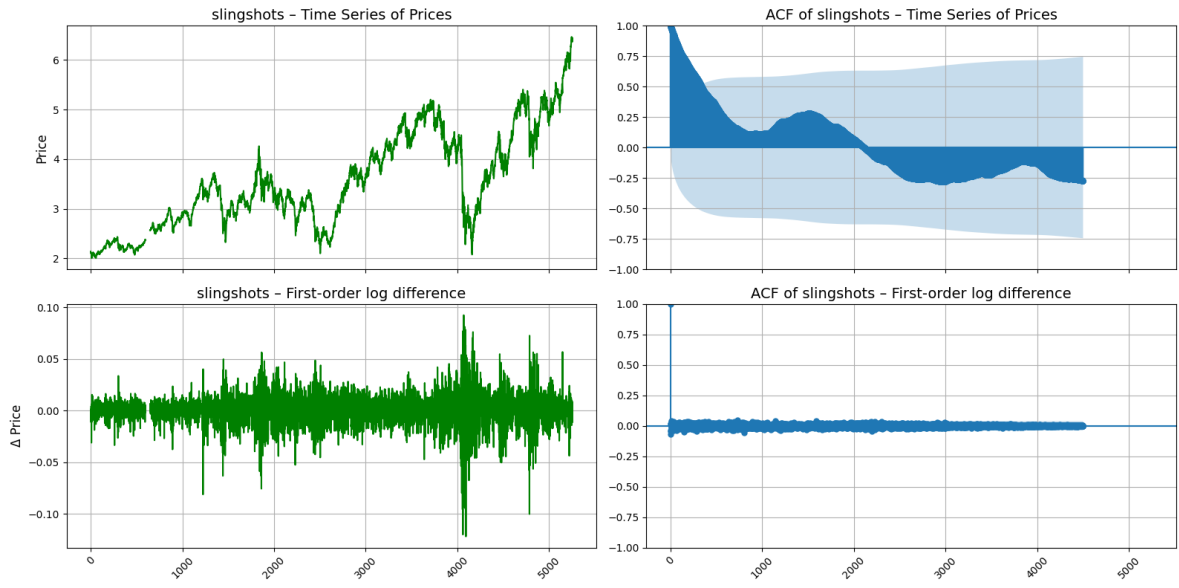


Figure 23: Time series of *slingshots* with corresponding ACFs and first-order log differences.

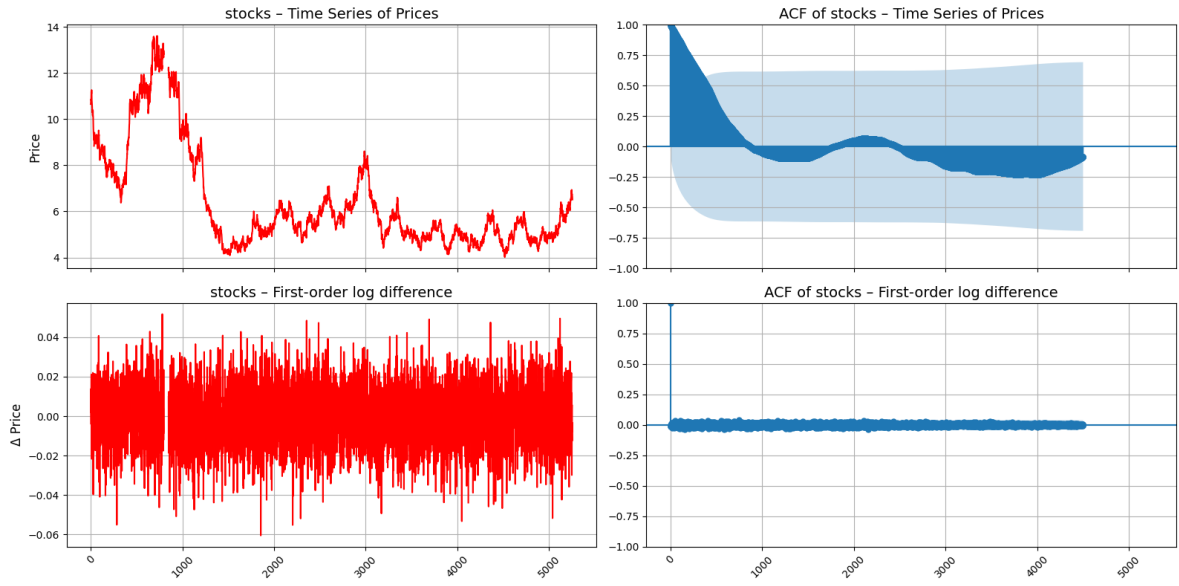


Figure 24: Time series of *stocks* with corresponding ACFs and first-order log differences.

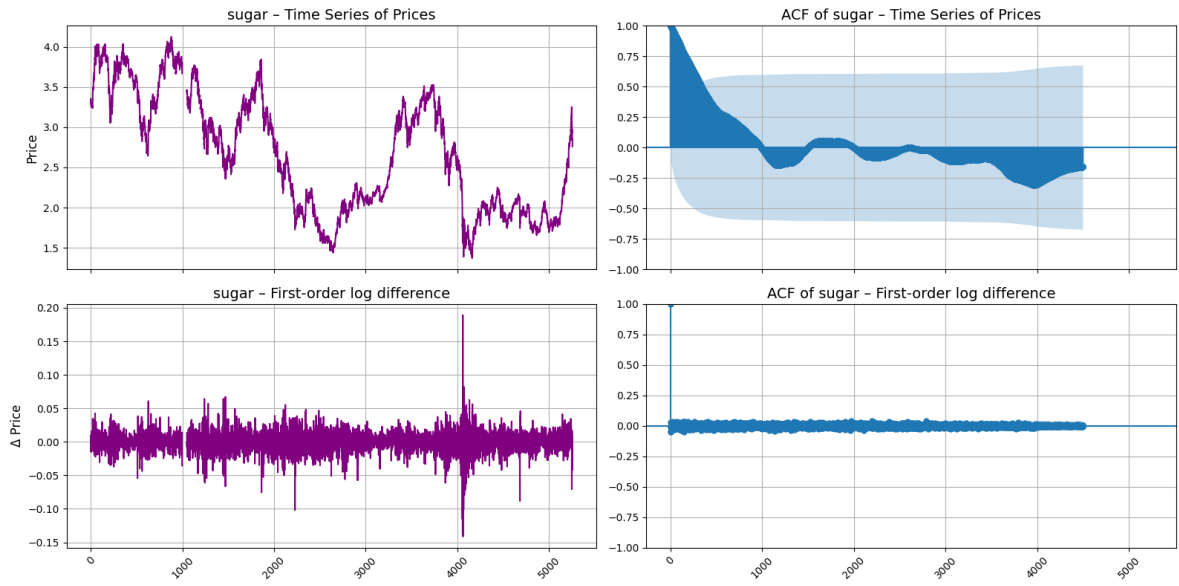


Figure 25: Time series of *sugar* with corresponding ACFs and first-order log differences.

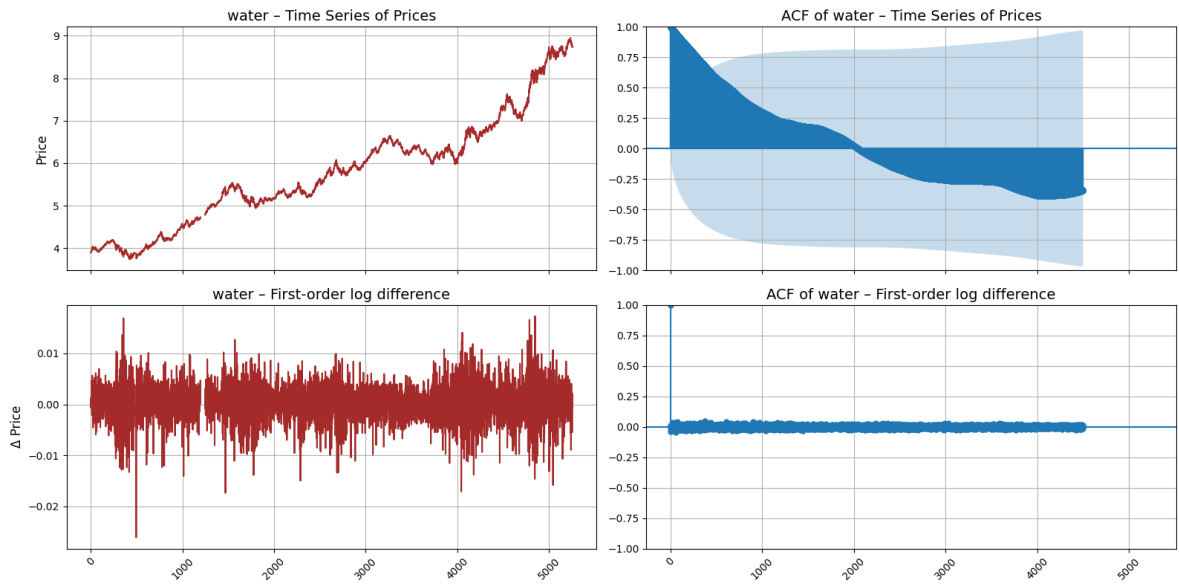


Figure 26: Time series of *water* with corresponding ACFs and first-order log differences.

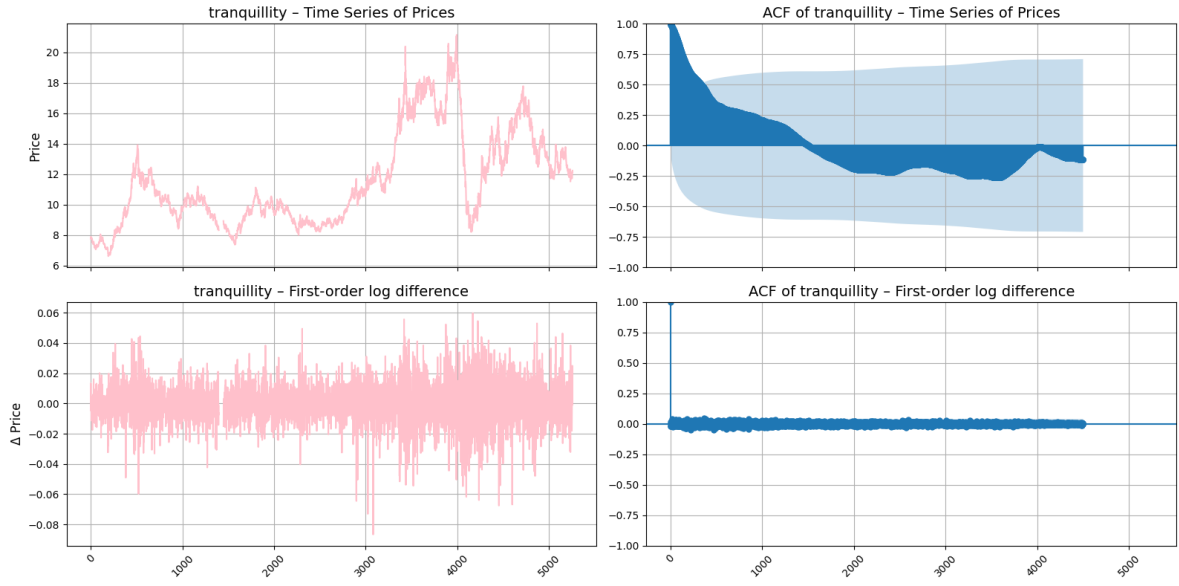


Figure 27: Time series of *tranquility* with corresponding ACFs and first-order log differences.

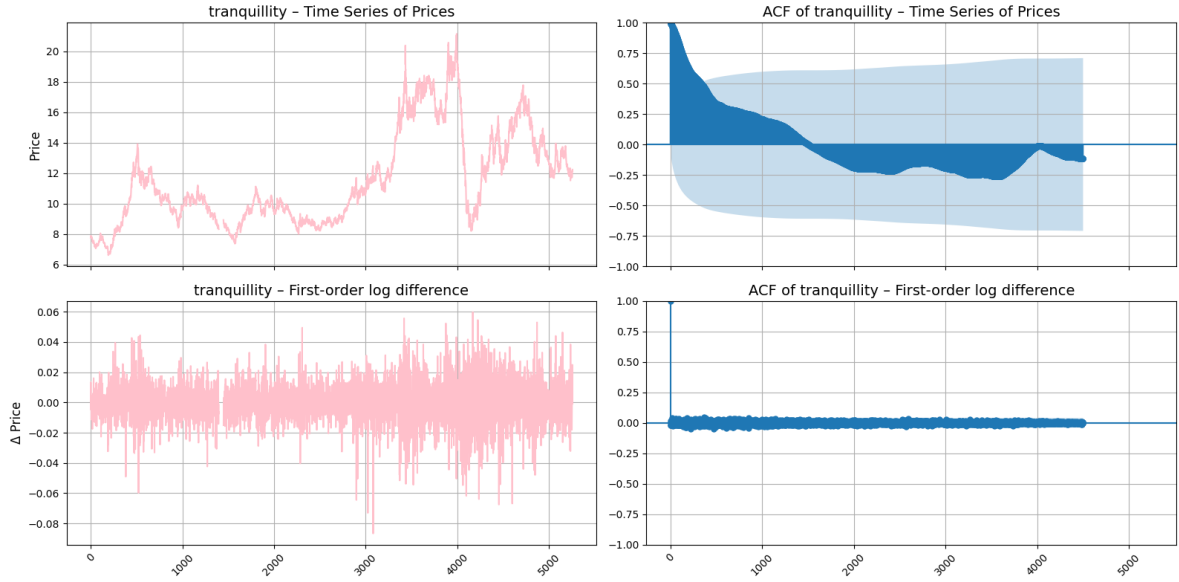


Figure 28: Time series of *tranquility* with corresponding ACFs and first-order log differences.

B.1 Seasonality

We tried several methods of trying to find seasonality, one method is called seasonality strength, it compares the variance of the seasonality and the rest of the data. This was done for 2 to 1000 days, it was found that 320-, 500-, and 855- day seasons had peaks. But when analyzing them, the amplitude was deemed low enough as to not bother with them.

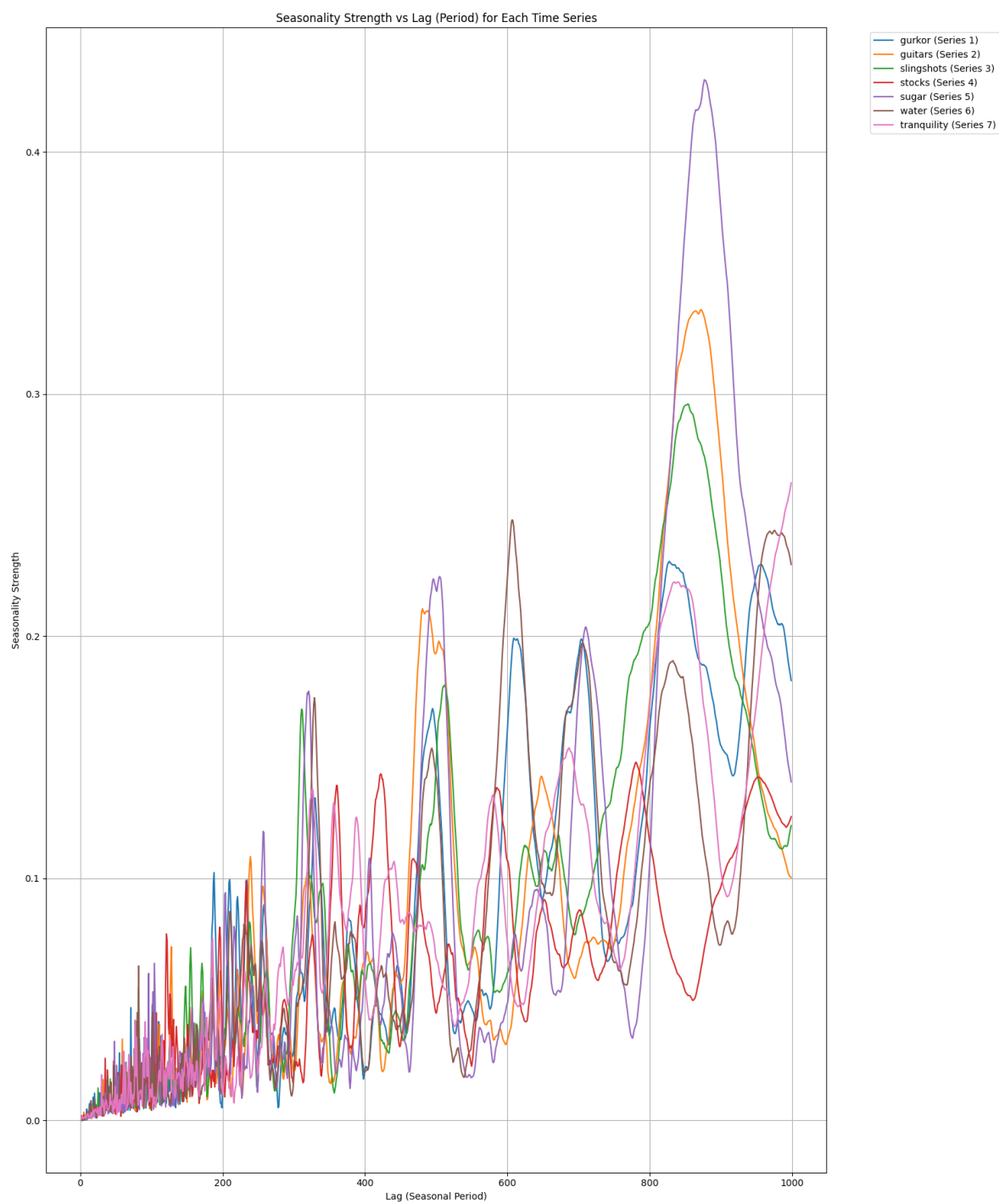


Figure 29: Seasonality strength of the time series.

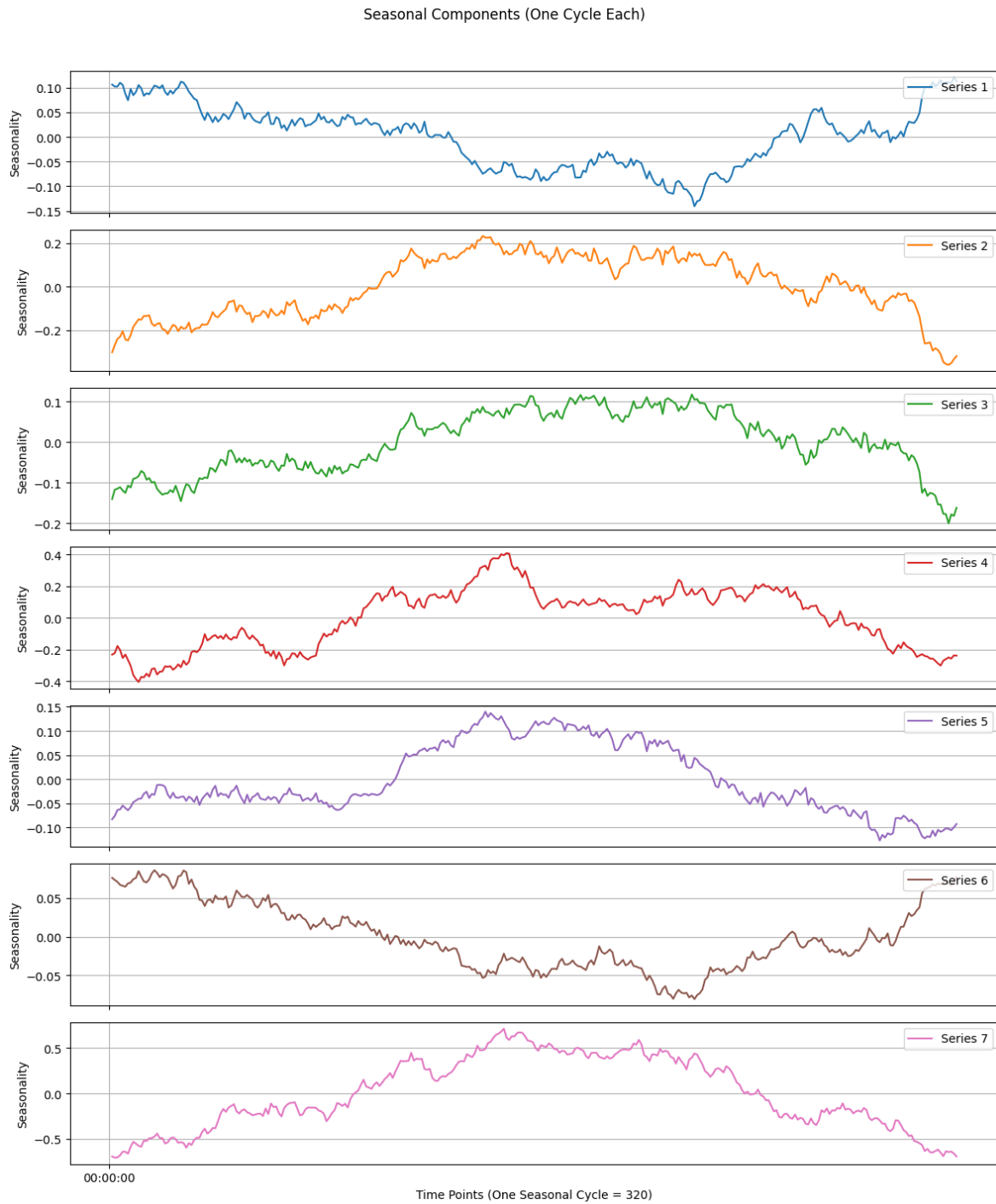


Figure 30: Seasonality strength of the time series.

B.2 Interpolation

All the interpolated data from Task 2.

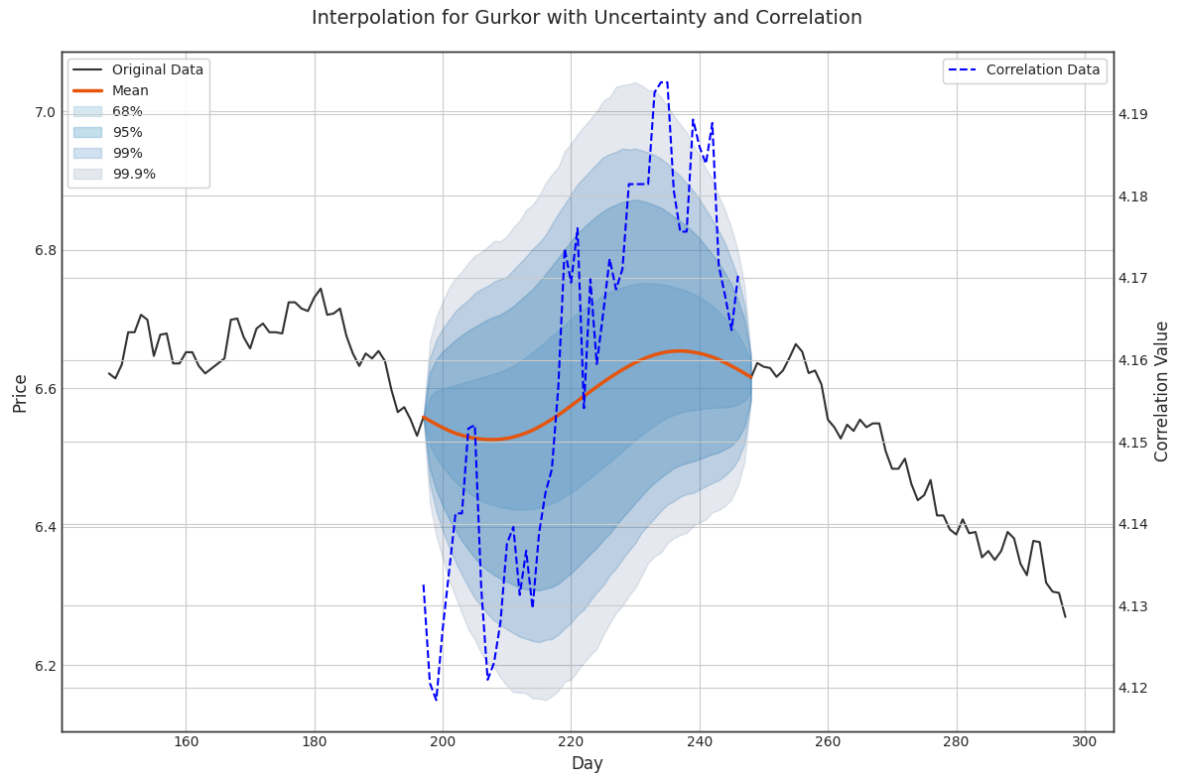


Figure 31: Interpolated time series of *Gurkor*.

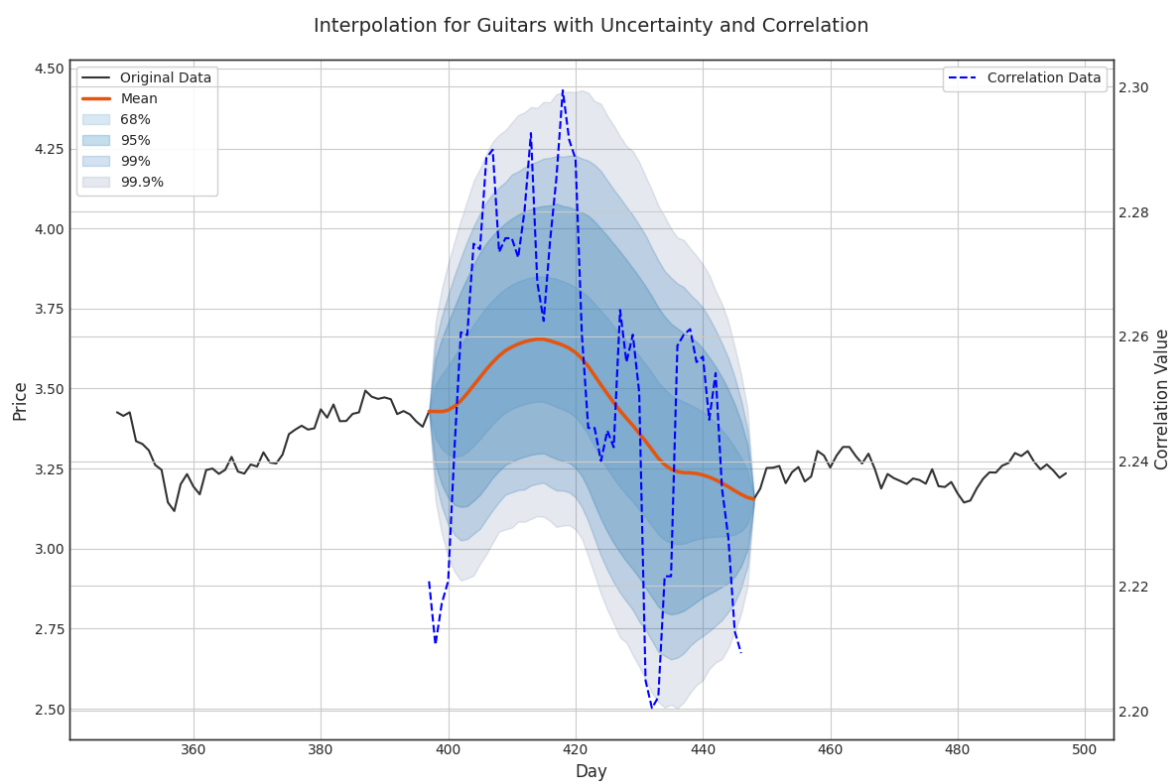


Figure 32: Interpolated time series of *Guitars*.

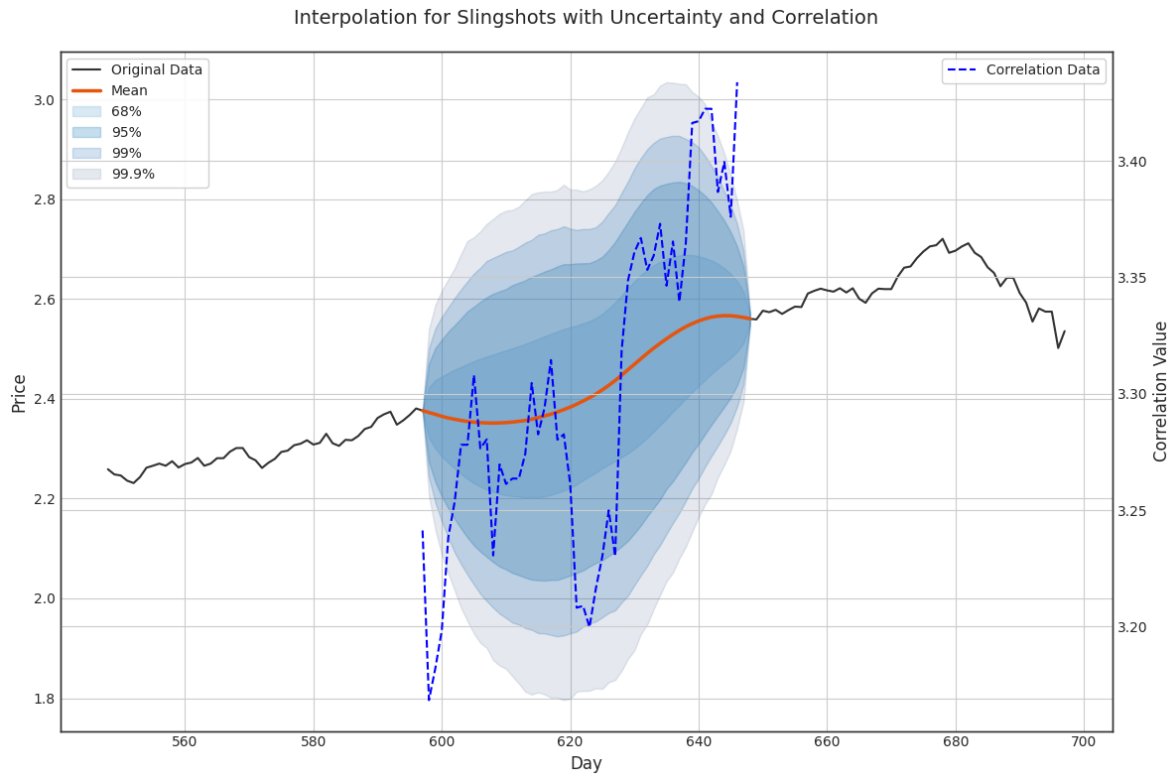


Figure 33: Interpolated time series of *Slingshots*.

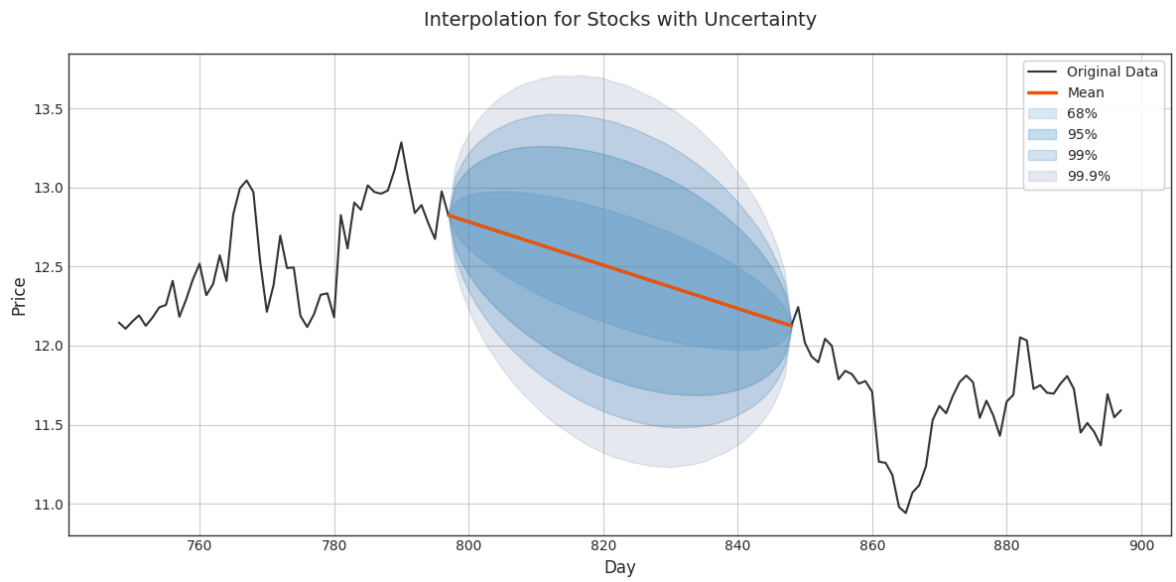


Figure 34: Interpolated time series of *Stocks*.

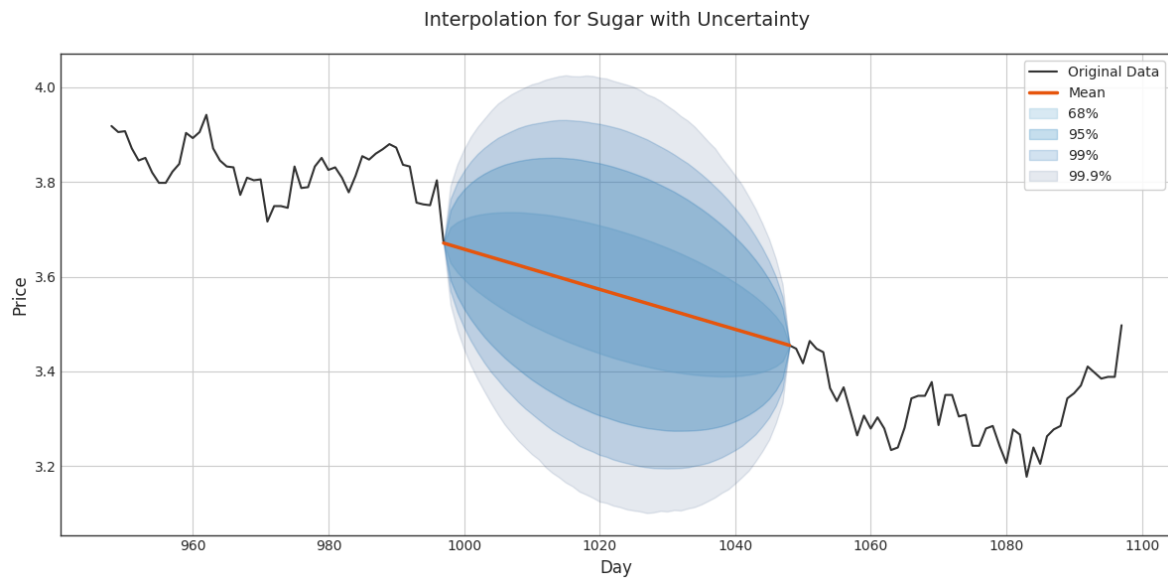


Figure 35: Interpolated time series of *Sugar*.

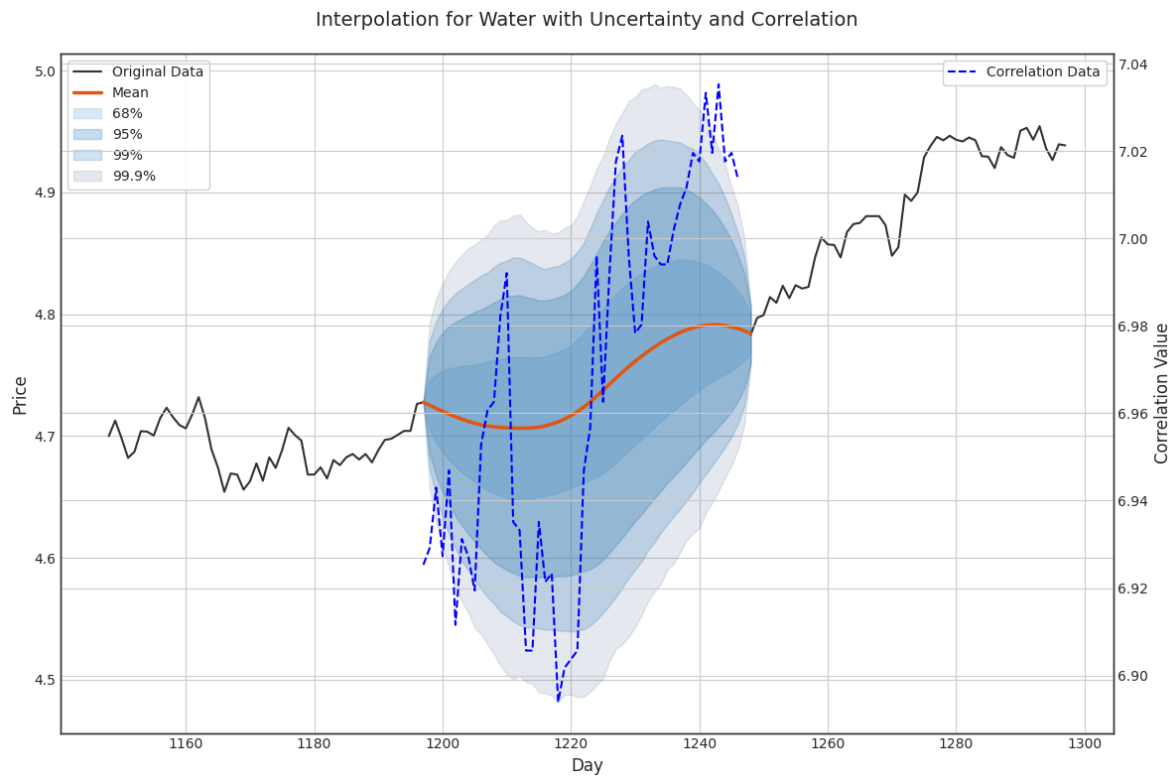


Figure 36: Interpolated time series of *Water*.

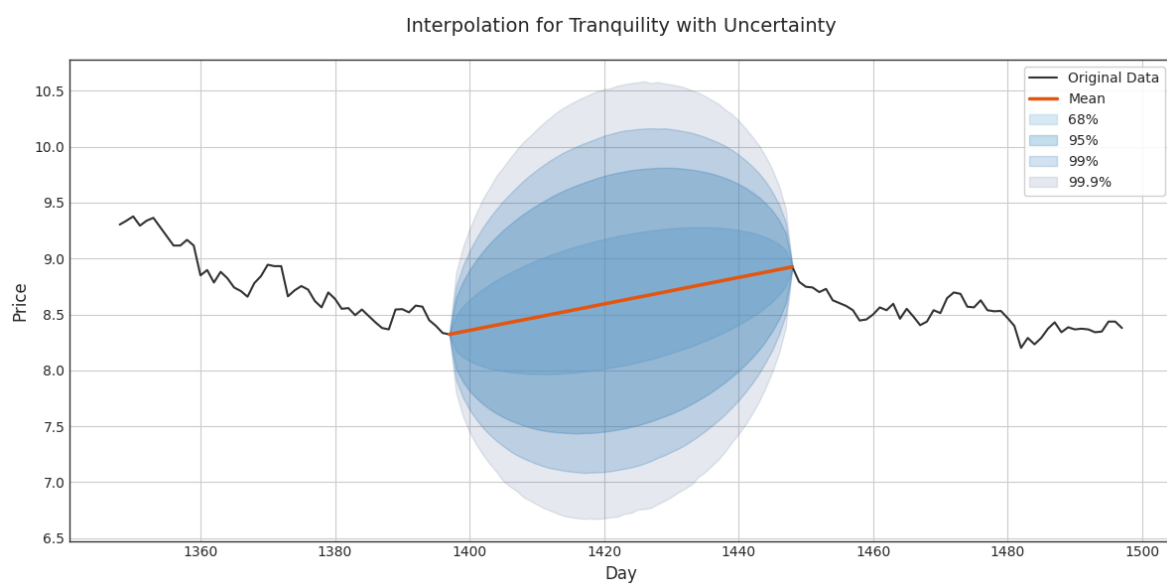


Figure 37: Interpolated time series of *Tranquility*.

Group contract/statement

We, the undersigned, certify that this report is entirely our work. We further confirm that all members of the group have contributed equally to the completion of this assignment.